共享單車需求預測：結合環境因素的分析

M132040013

研究目的

本研究旨在深入探討環境因素（如氣候條件、季節變化、地理特徵等）如何共同影響共享單車的需求模式。隨著共享單車在全球城市交通系統中逐漸普及，理解這些因素對需求波動的影響，對於共享單車的運營優化和資源配置具有重要意義。

變數介紹

目標變數:cnt(每小時共享單車使用數量)

考慮先丟棄時間變數dteday與instant(原有ID)去進行資料探勘

資料筆數:17379, 缺失值:0

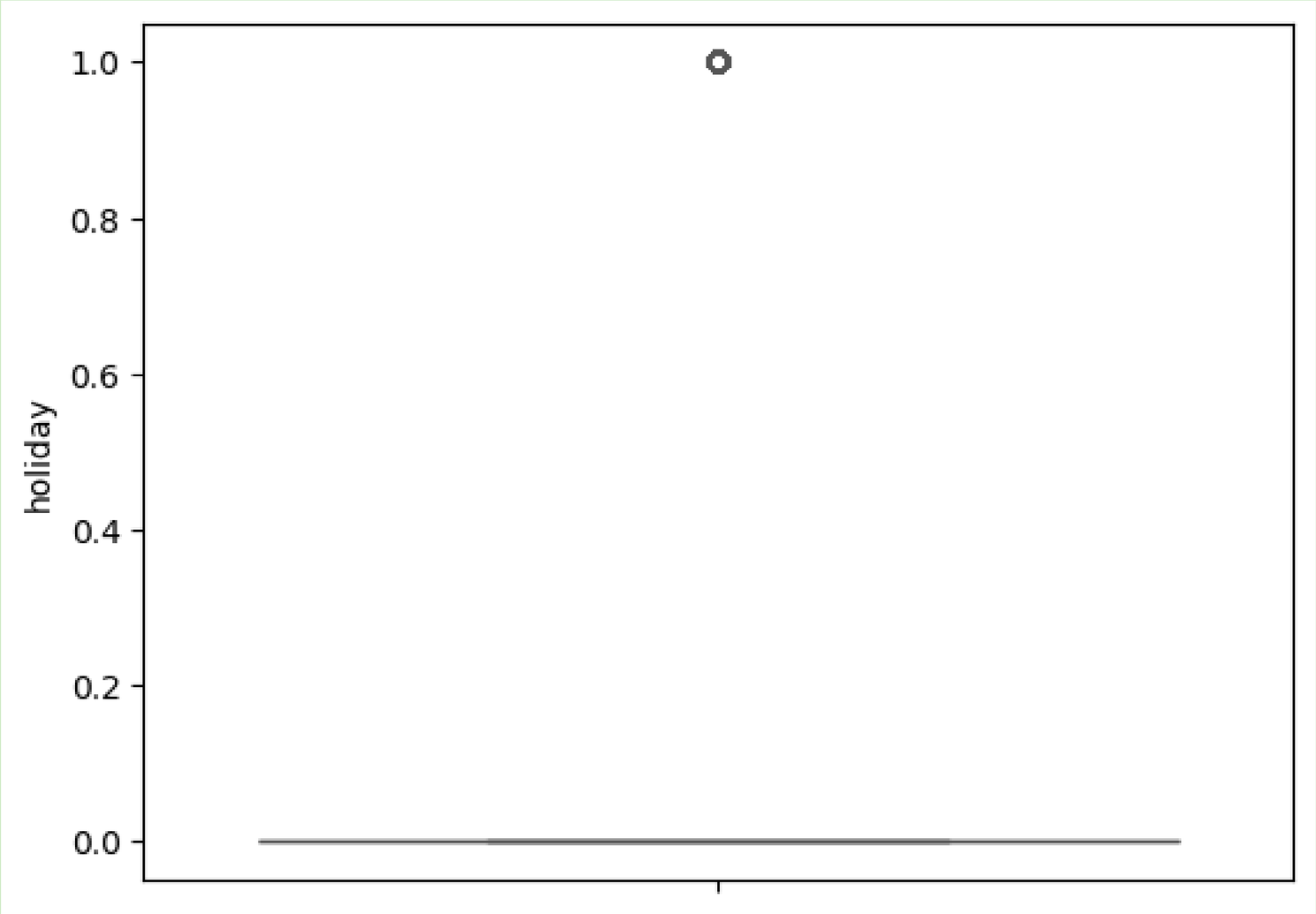
特徵數量:15

類別變數

season	yr	mnth	hr	holiday	week day	working day	weather sit
1:冬季	0: 2011年	月份 (1~12): 對應 1~12月	小時 (0~23): 對應 0~23時	0: 非假期	0~6: 對應 星期幾	0: 非工作日 (六日)	1~4: 1:晴天
2:春季	1: 2012年			1: 假期	0:星期日	1: 工作日	2:少雲
3:夏季							3:多雲
4:秋季							4:陰天

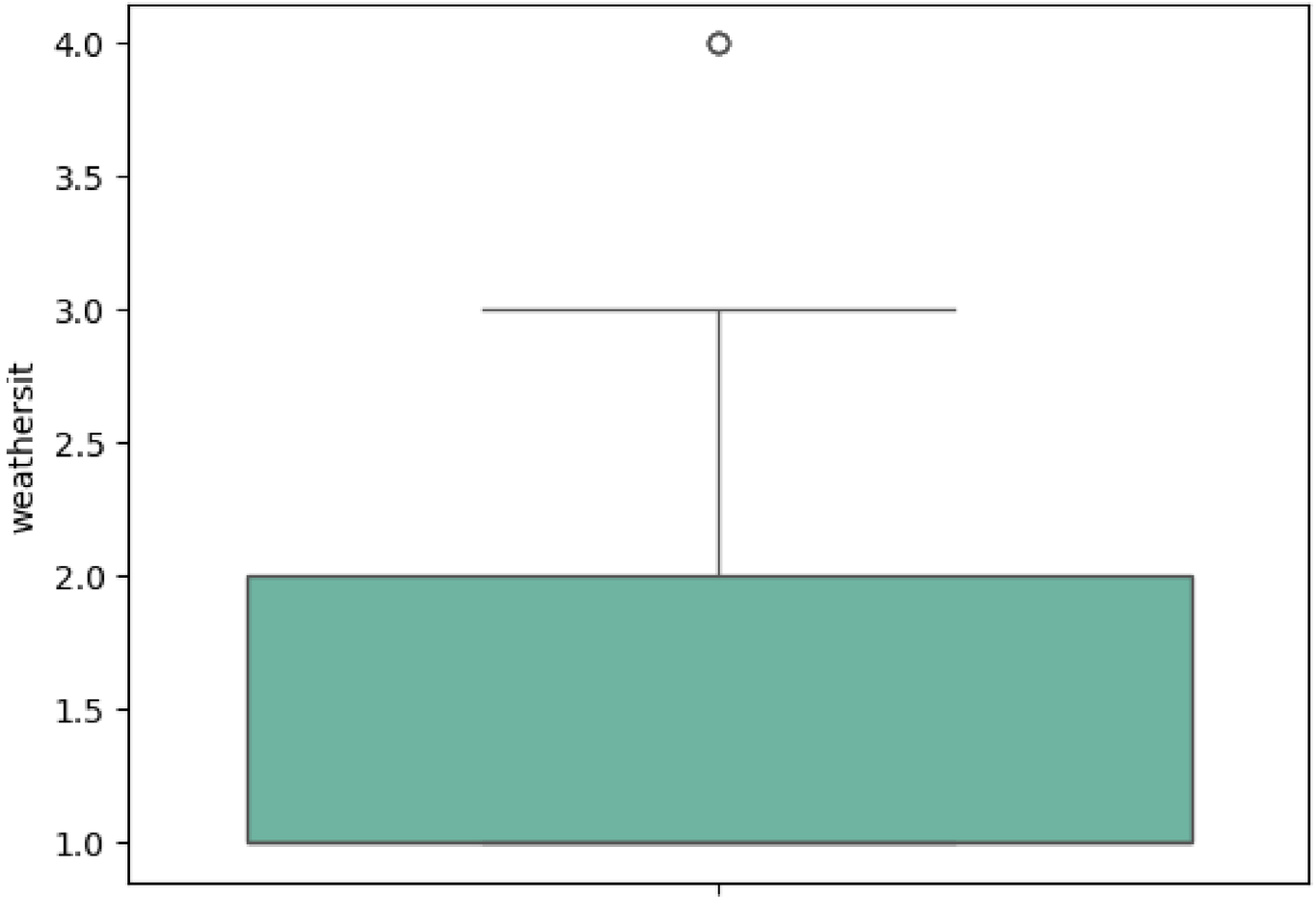
數值類	temp	atemp	hum	windspeed	casual	registered	cnt (目標)
mean	0.497	0.476	0.627	0.19	35.676	153.787	189.463
std	0.192	0.172	0.193	0.122	49.305	151.357	181.388
min	0.02	0	0	0	0	0	1
max	1	1	1	0.85	367	886	977

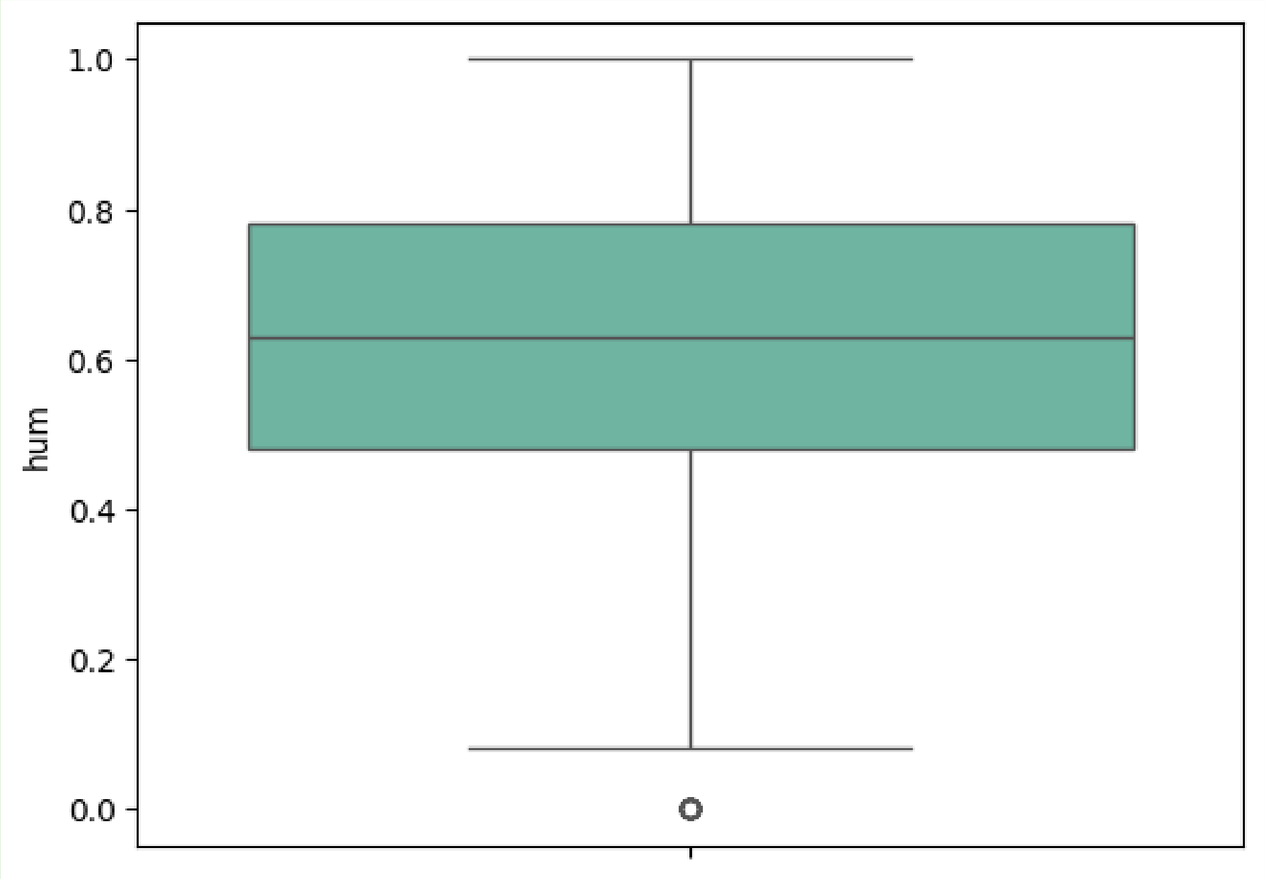
盒鬚圖



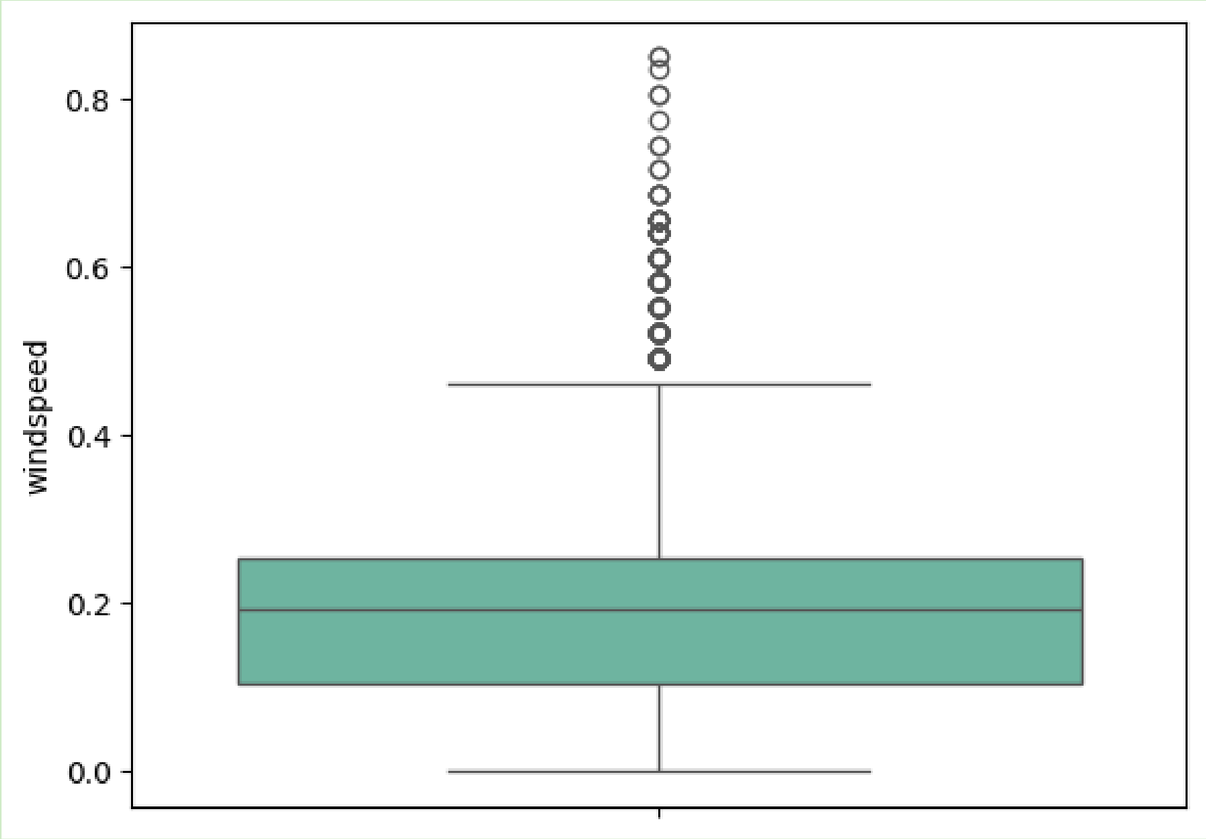
dteday	count
2011-01-17	24
2012-11-22	24
2012-07-04	24
2012-05-28	24
2012-04-16	24
2012-11-12	24
2012-01-16	24
2011-02-21	24
2011-11-24	24
2012-10-08	24
2011-11-11	24
2011-10-10	24
2011-09-05	24
2011-07-04	24
2011-05-30	24
2011-04-15	24
2012-09-03	24
2012-01-02	23
2012-02-20	23
2011-12-26	23
2012-12-25	23

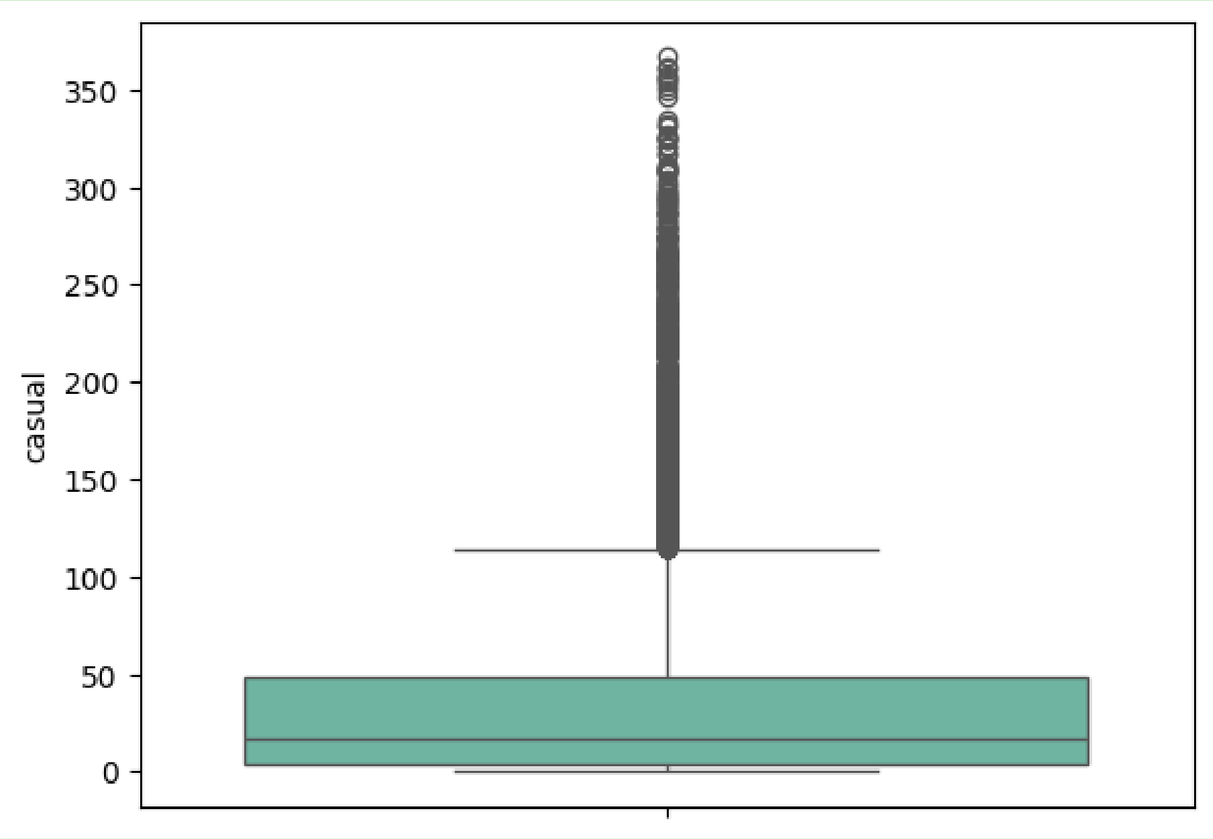
	instant	dteday	season	yr	mnth	hr	holiday	weekday	workingday	weathersit
585	586	2011-01-26	1	0	1	16	0	3	1	4
8854	8855	2012-01-09	1	1	1	18	0	1	1	4
9123	9124	2012-01-21	1	1	1	1	0	6	0	4

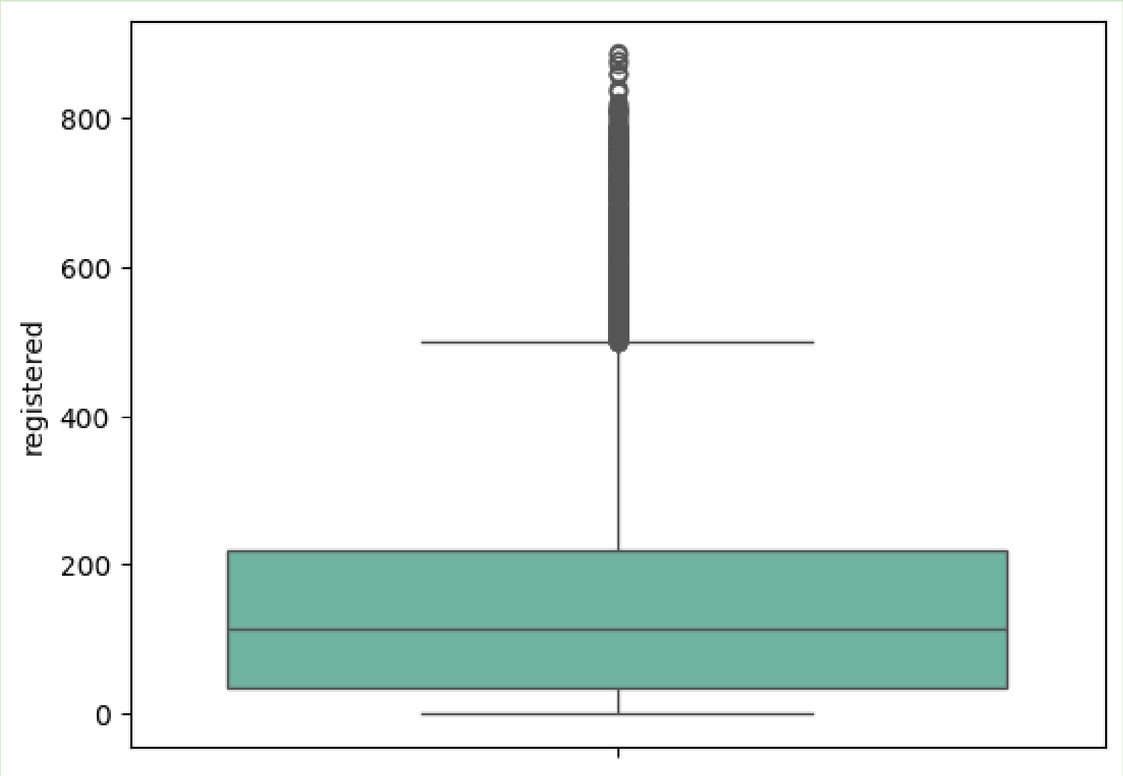


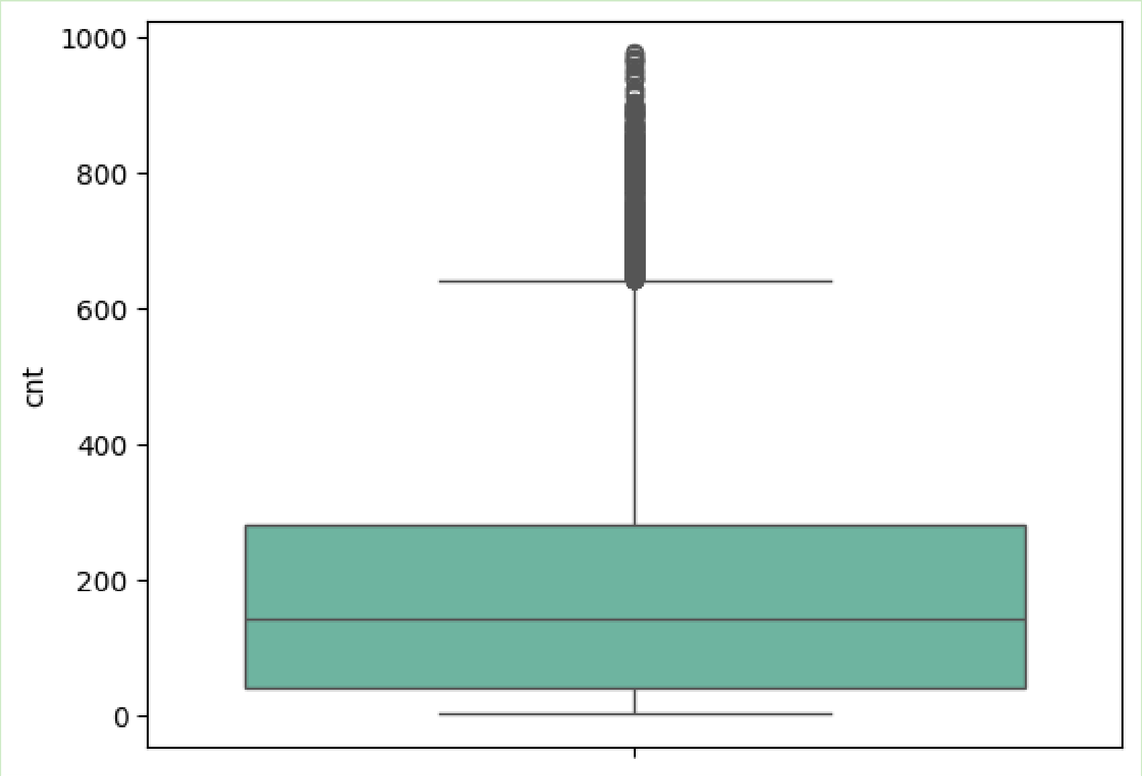


instant	dteday	season	yr	mnth	hr	holiday	weekday	workingday	weathersit
1552	2011-03-10	1	0	3	0	0	4	1	3
1553	2011-03-10	1	0	3	1	0	4	1	3
1572	2011-03-10	1	0	3	22	0	4	1	2
1571	2011-03-10	1	0	3	21	0	4	1	3
1570	2011-03-10	1	0	3	20	0	4	1	3
1569	2011-03-10	1	0	3	19	0	4	1	3
1568	2011-03-10	1	0	3	18	0	4	1	3
1567	2011-03-10	1	0	3	17	0	4	1	2
1566	2011-03-10	1	0	3	16	0	4	1	3
1565	2011-03-10	1	0	3	15	0	4	1	3
1564	2011-03-10	1	0	3	14	0	4	1	3
1563	2011-03-10	1	0	3	13	0	4	1	3
1562	2011-03-10	1	0	3	12	0	4	1	3
1561	2011-03-10	1	0	3	11	0	4	1	3
1560	2011-03-10	1	0	3	10	0	4	1	3
1559	2011-03-10	1	0	3	9	0	4	1	3
1558	2011-03-10	1	0	3	8	0	4	1	3
1557	2011-03-10	1	0	3	7	0	4	1	3
1556	2011-03-10	1	0	3	6	0	4	1	3
1555	2011-03-10	1	0	3	5	0	4	1	3
1554	2011-03-10	1	0	3	2	0	4	1	3
1573	2011-03-10	1	0	3	23	0	4	1	3

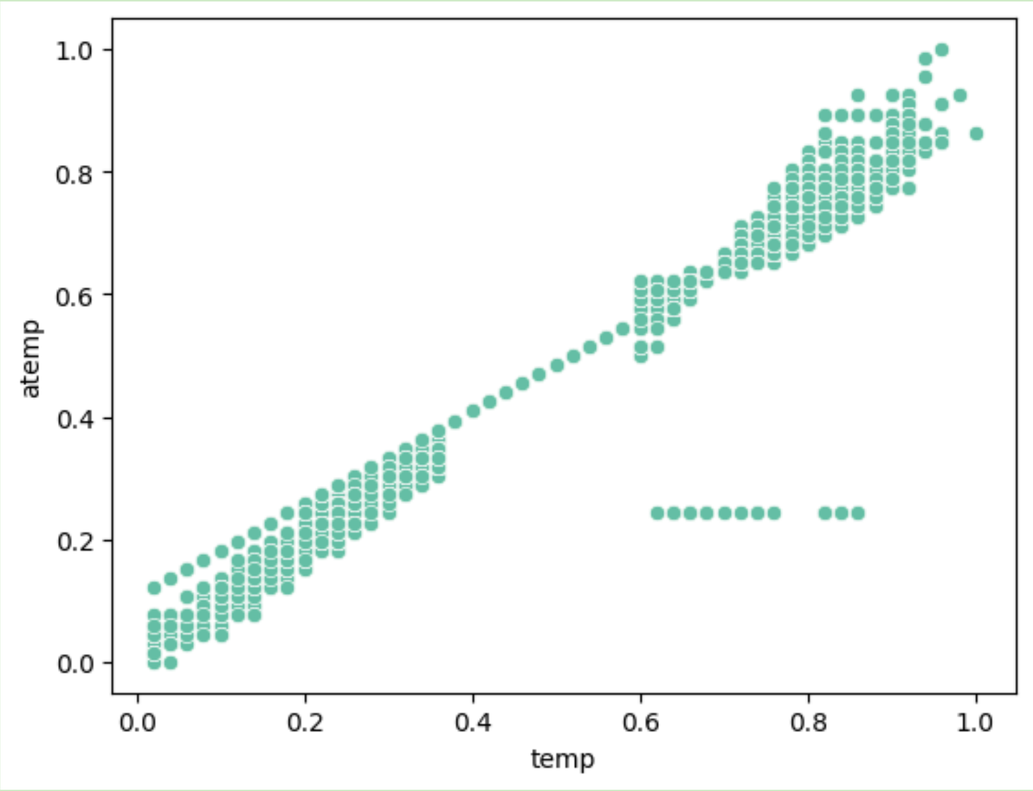

windspeed
0.0
0.2
0.4
0.6
0.8

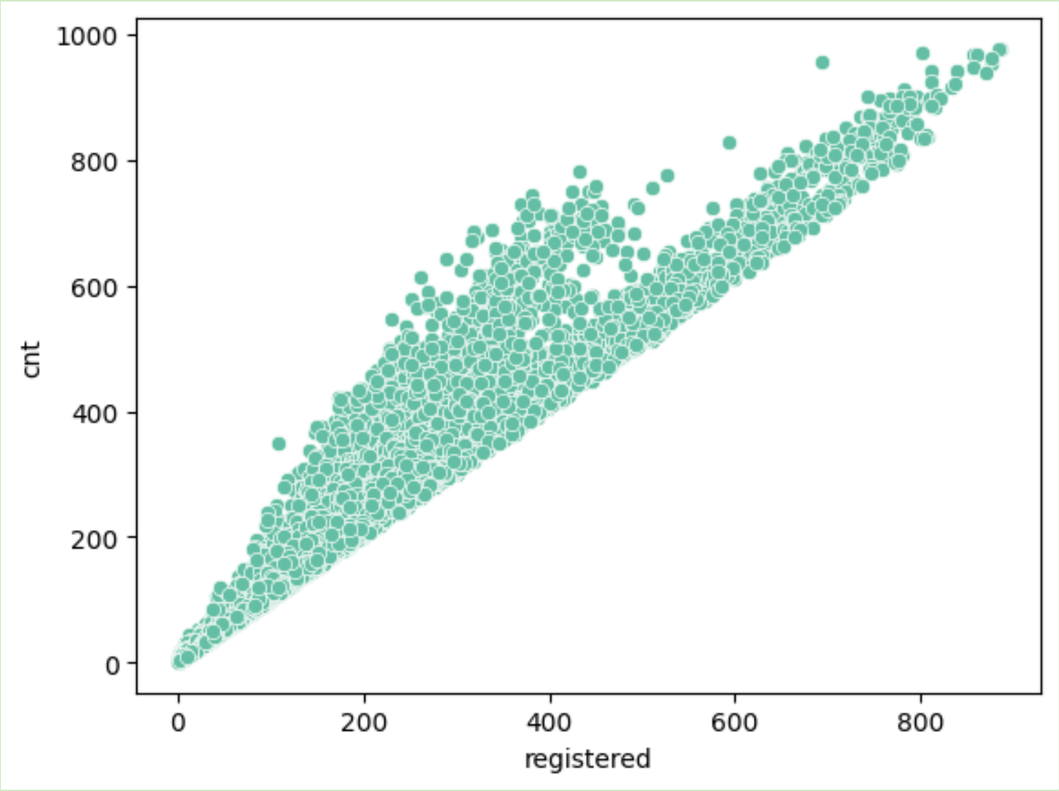

casual
0
50
100
150
200
250
300
350

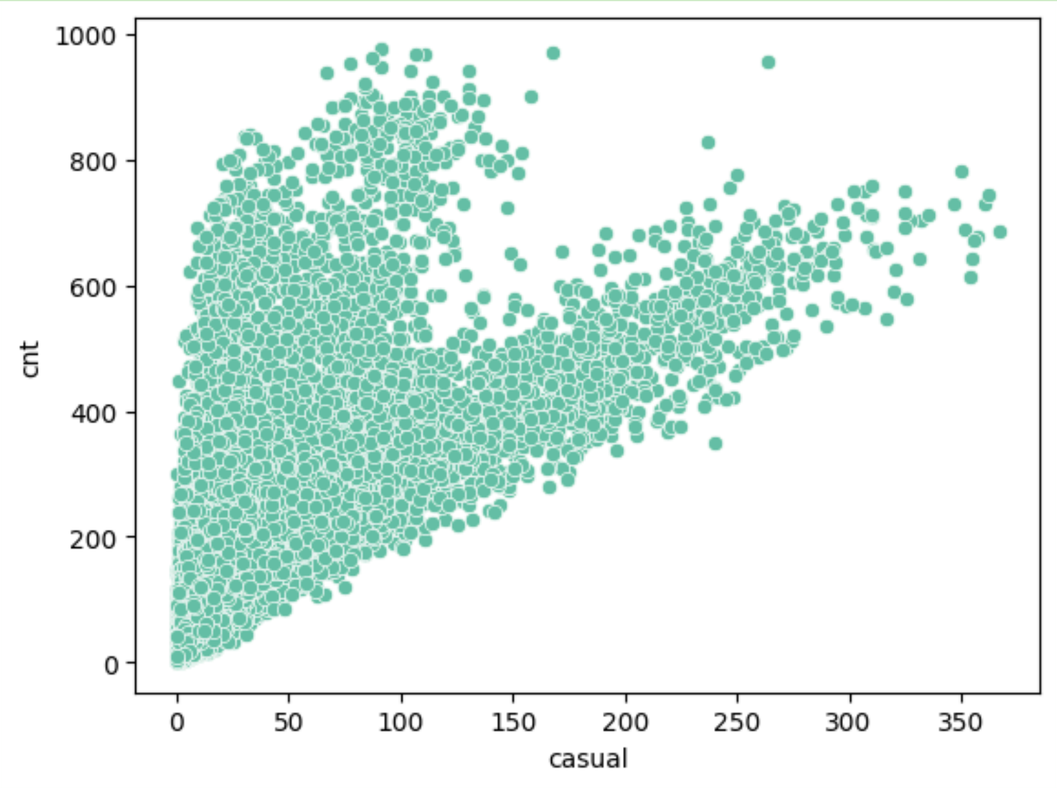

registered
0
200
400
600
800

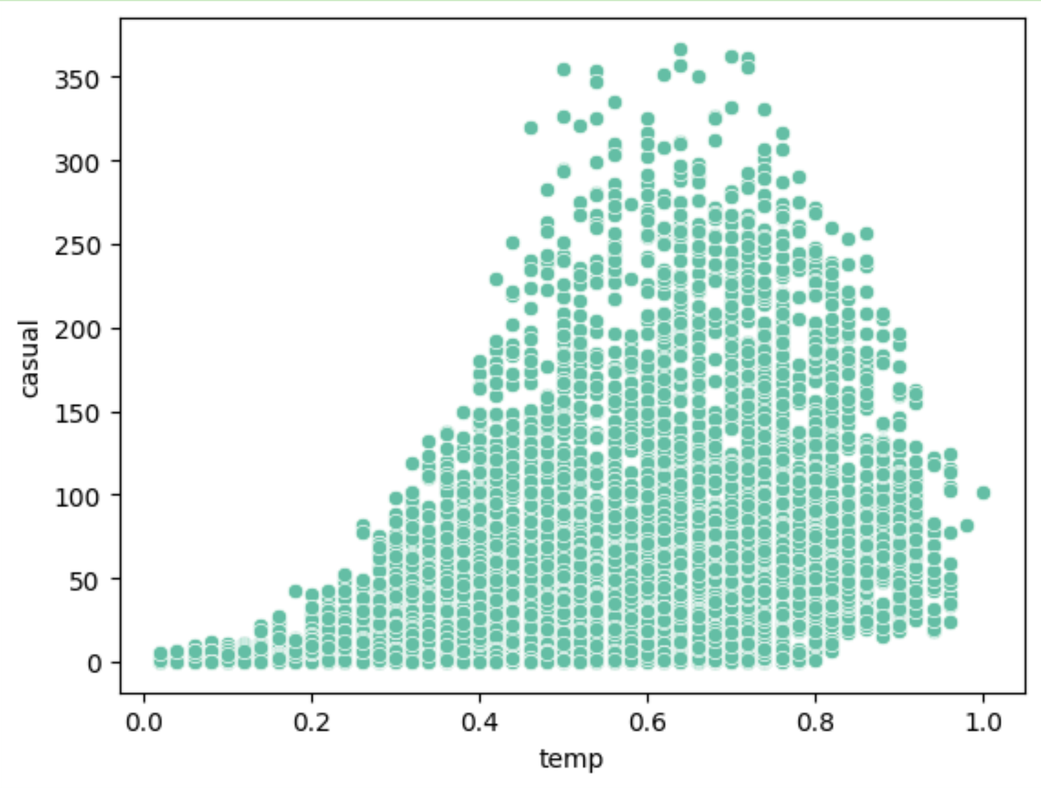

cnt
0
200
400
600
800
1000

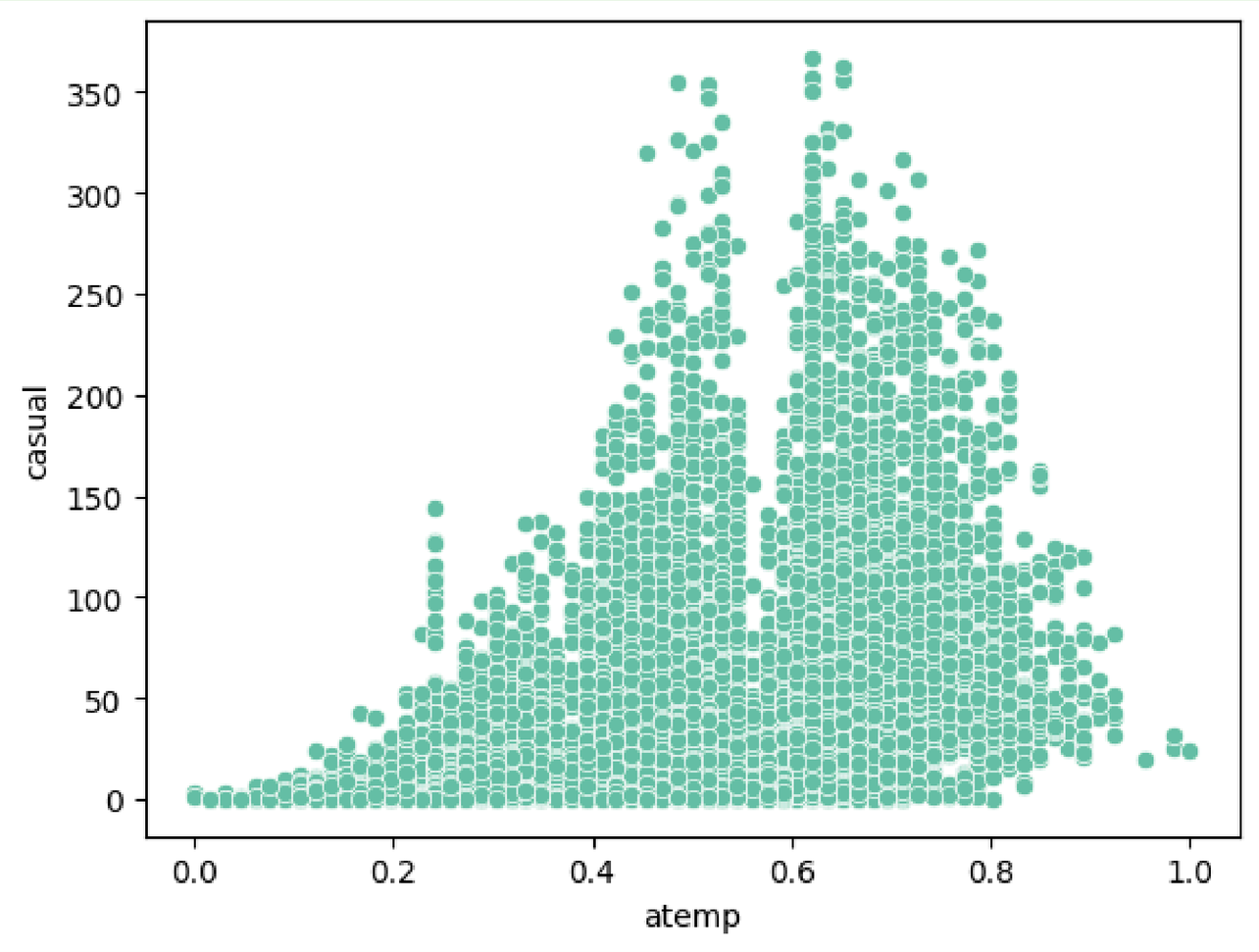
ScatterPlot



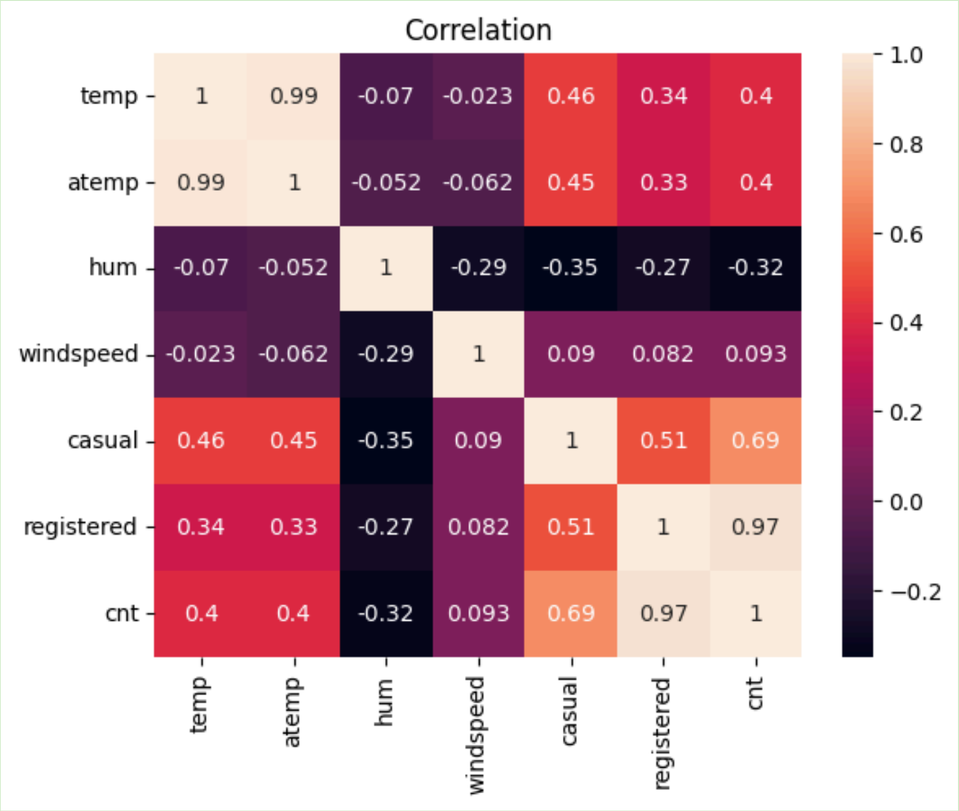


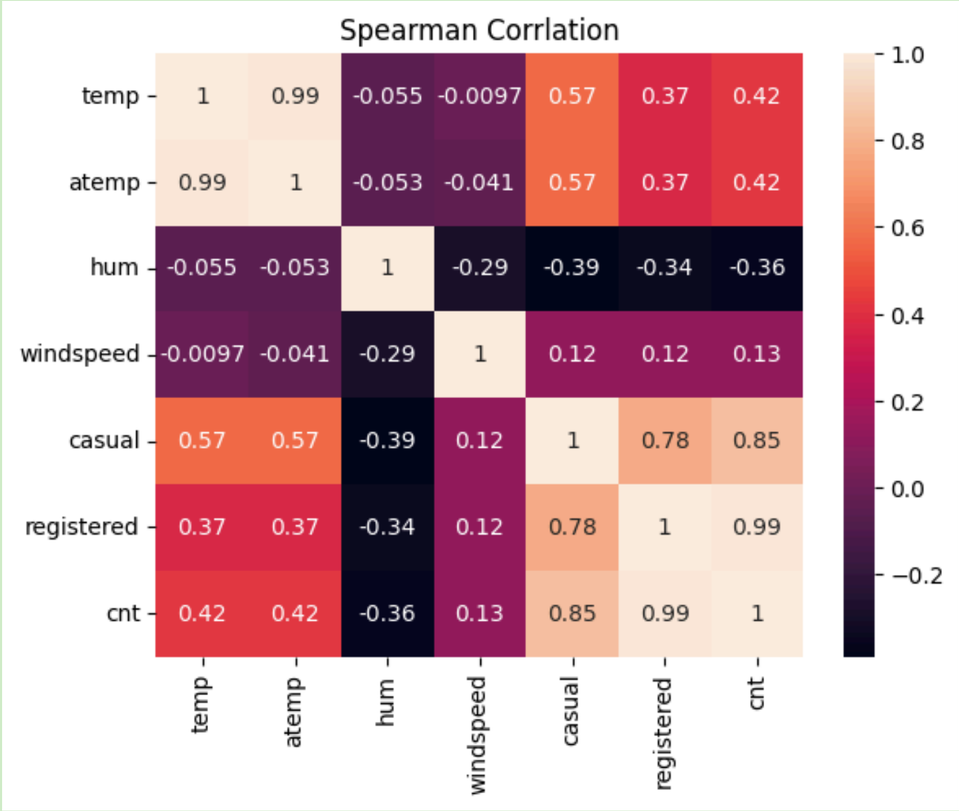

1000
800
600
400
200
0
cnt
0
50
100
150
200
250
300
350
casual


350
300
250
200
150
100
50
0
casual
0.0
0.2
0.4
0.6
0.8
1.0
temp

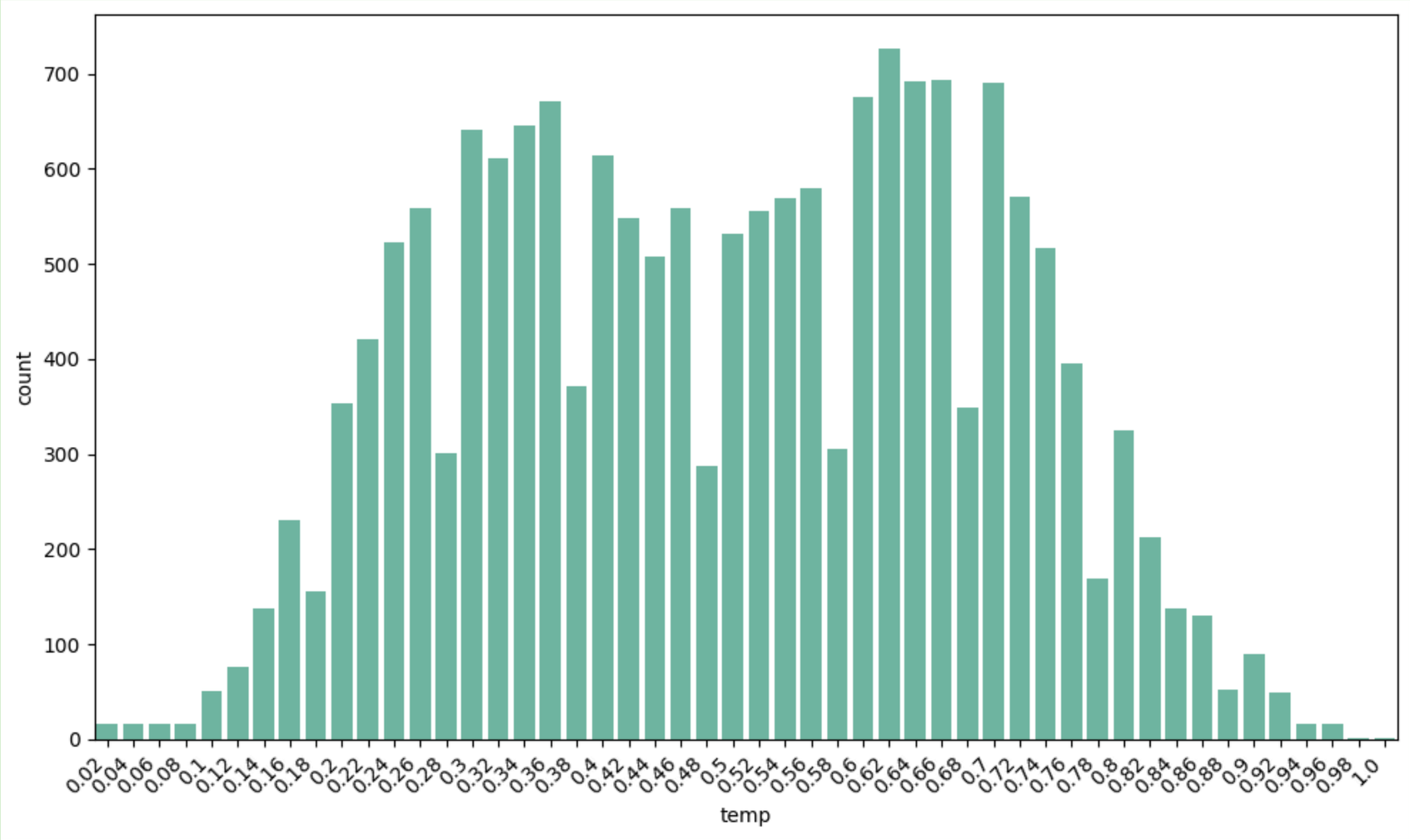

350
300
250
200
150
100
50
0
casual
0.0
0.2
0.4
0.6
0.8
1.0
atemp

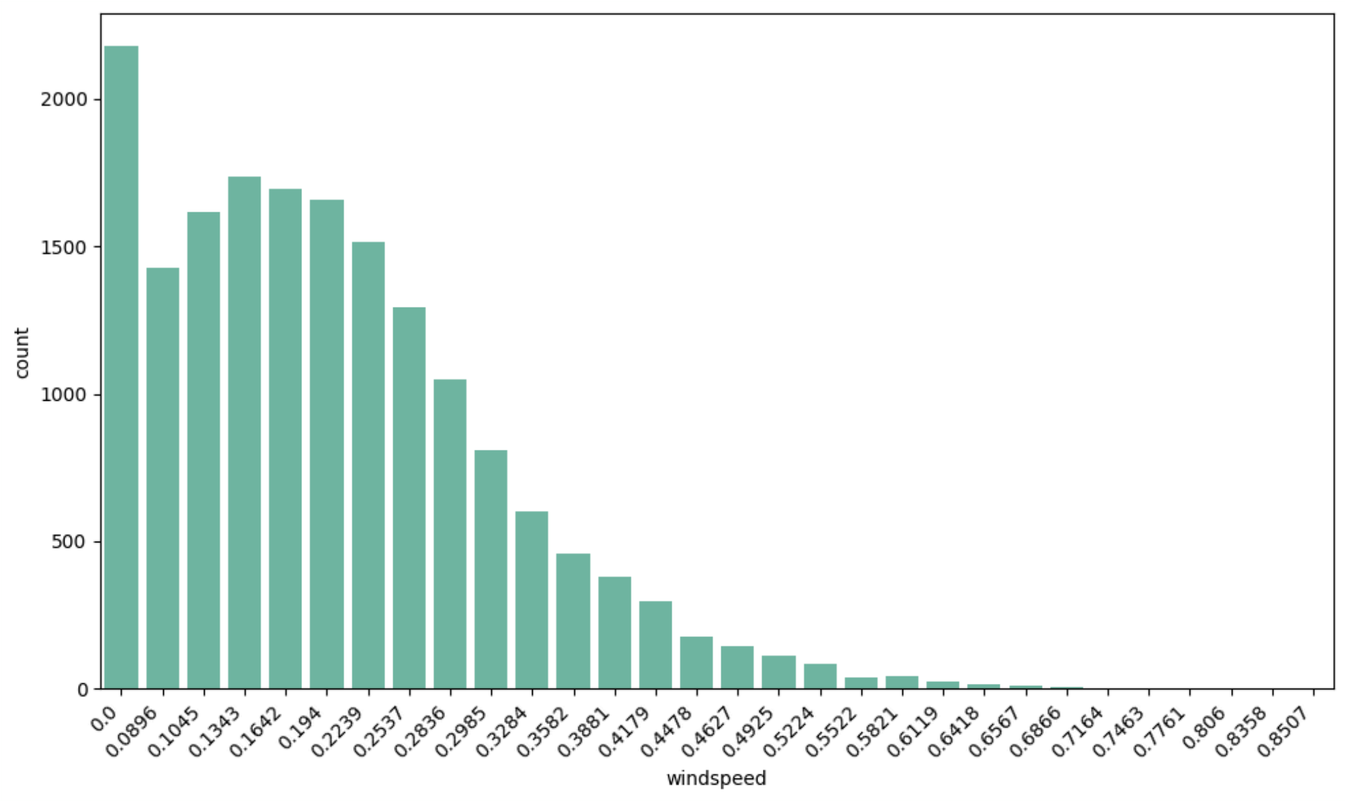
相關係數

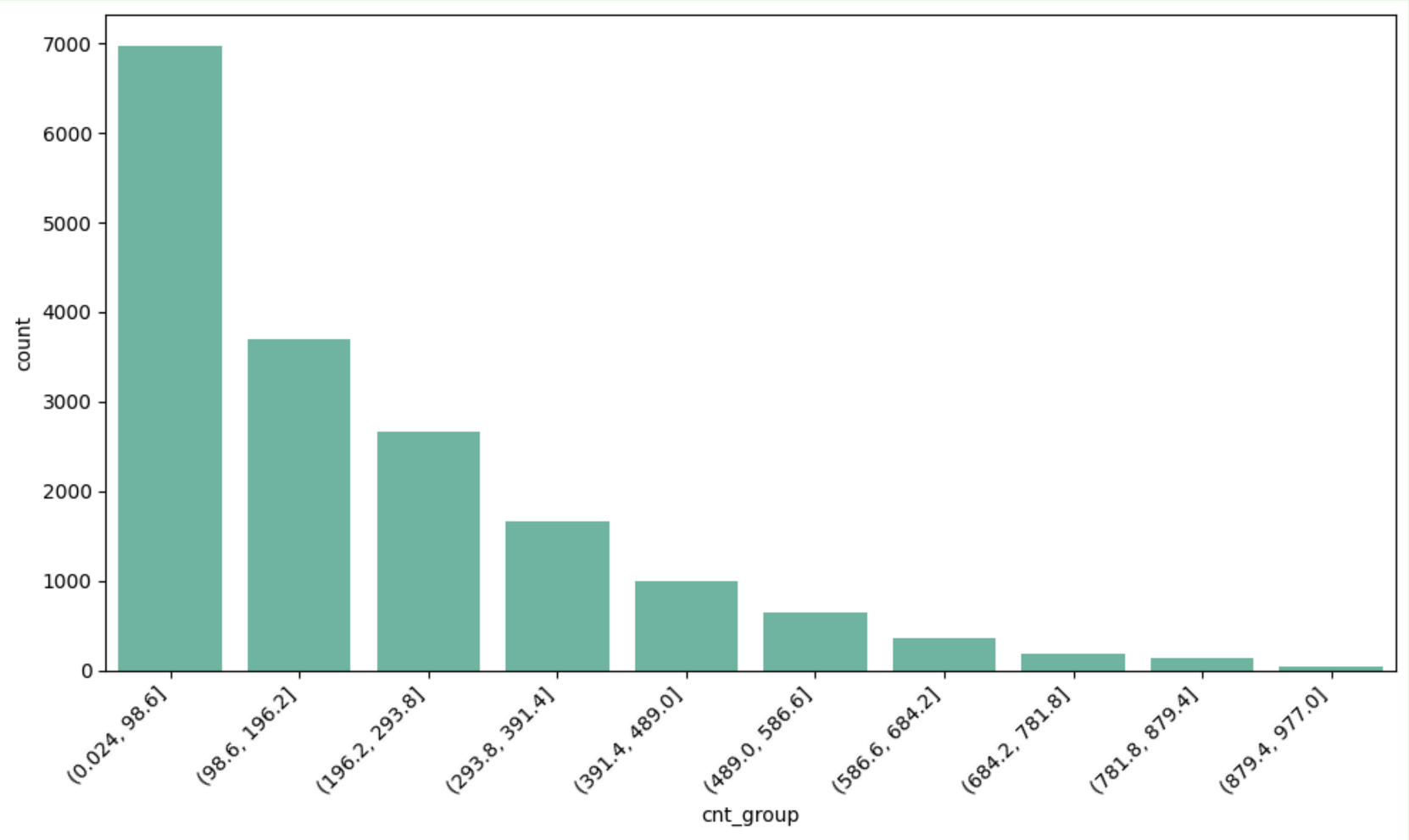

Correlation
temp 1 0.99 -0.07 -0.023 0.46 0.34 0.4
atemp 0.99 1 -0.052 -0.062 0.45 0.33 0.4
hum -0.07 -0.052 1 -0.29 -0.35 -0.27 -0.32
windspeed -0.023 -0.062 -0.29 1 0.09 0.082 0.093
casual 0.46 0.45 -0.35 0.09 1 0.51 0.69
registered 0.34 0.33 -0.27 0.082 0.51 1 0.97
cnt 0.4 0.4 -0.32 0.093 0.69 0.97 1
temp atemp hum windspeed casual registered cnt
1.0 0.8 0.6 0.4 0.2 0.0 −0.2


Spearman Corrlation
temp 1 0.99 -0.055 -0.0097 0.57 0.37 0.42
atemp 0.99 1 -0.053 -0.041 0.57 0.37 0.42
hum -0.055 -0.053 1 -0.29 -0.39 -0.34 -0.36
windspeed -0.0097 -0.041 -0.29 1 0.12 0.12 0.13
casual 0.57 0.57 -0.39 0.12 1 0.78 0.85
registered 0.37 0.37 -0.34 0.12 0.78 1 0.99
cnt 0.42 0.42 -0.36 0.13 0.85 0.99 1
temp atemp hum windspeed casual registered cnt
1.0 0.8 0.6 0.4 0.2 0.0 −0.2

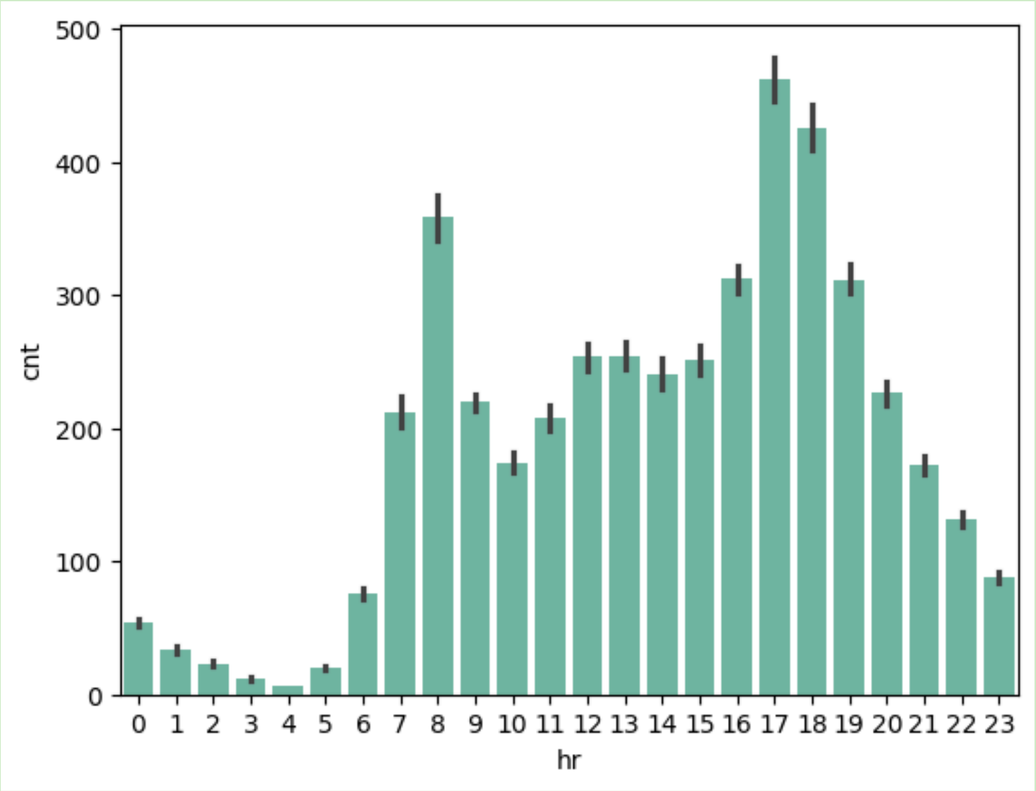
分布圖

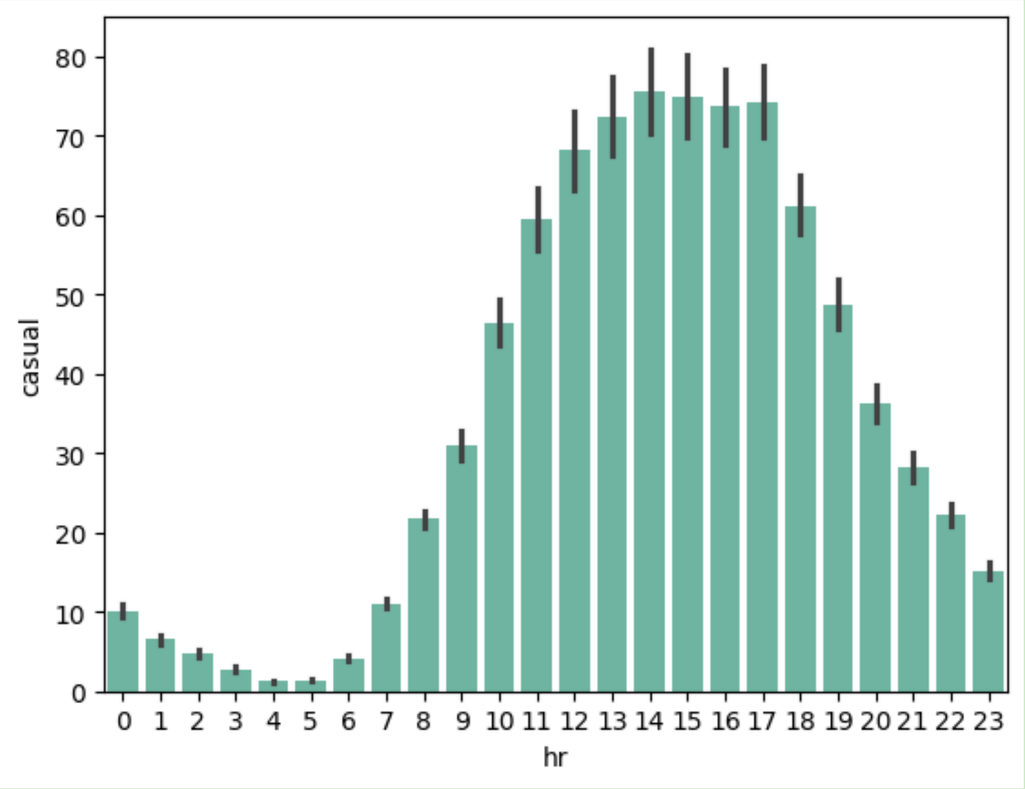

count
0
100
200
300
400
500
600
700
temp
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.24
0.26
0.28
0.3
0.32
0.34
0.36
0.38
0.4
0.42
0.44
0.46
0.48
0.5
0.52
0.54
0.56
0.58
0.6
0.62
0.64
0.66
0.68
0.7
0.72
0.74
0.76
0.78
0.8
0.82
0.84
0.86
0.88
0.9
0.92
0.94
0.96
0.98
1.0

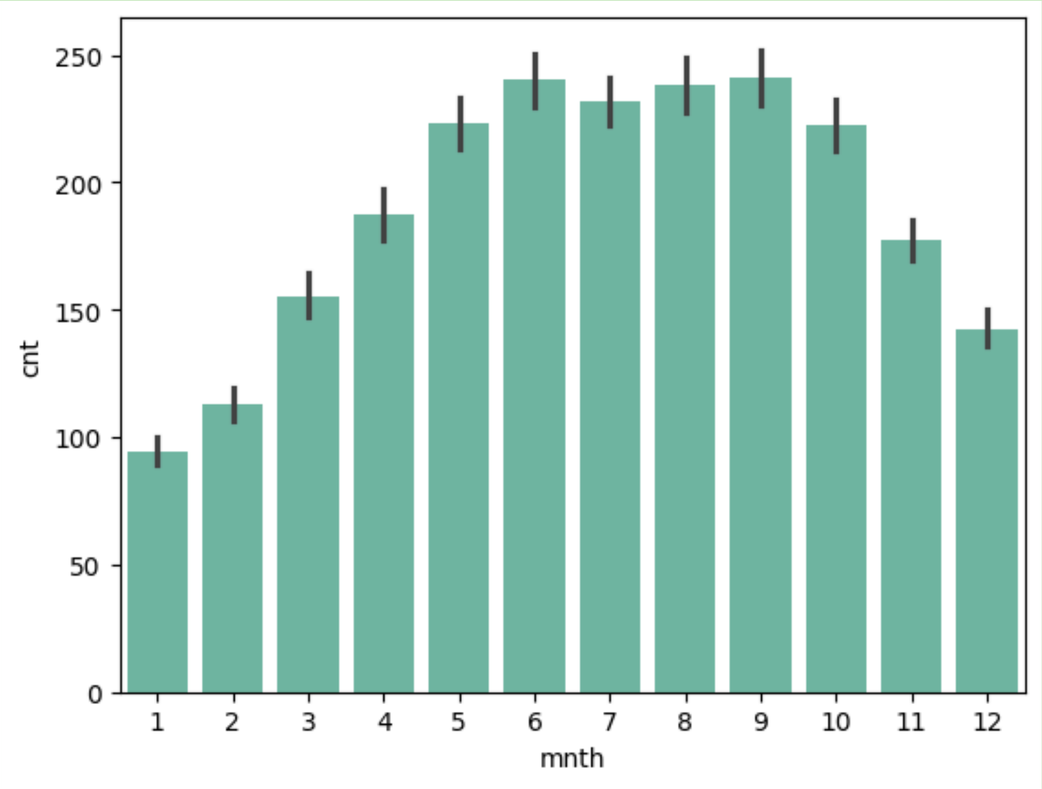

count
2000
1500
1000
500
0
0.0
0.0896
0.1045
0.1343
0.1642
0.194
0.2239
0.2537
0.2836
0.2985
0.3284
0.3582
0.3881
0.4179
0.4478
0.4627
0.4925
0.5224
0.5522
0.5821
0.6119
0.6418
0.6567
0.6866
0.7164
0.7463
0.7761
0.806
0.8358
0.8507
windspeed

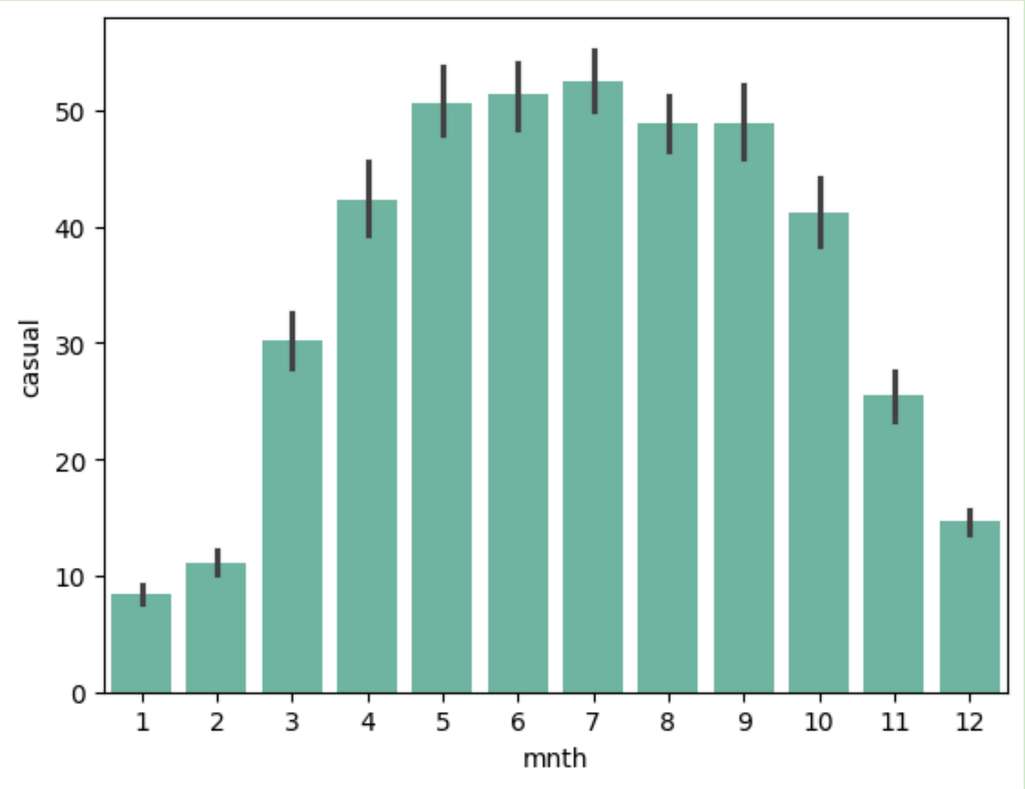

7000
6000
5000
4000
3000
2000
1000
0
count
(0.024, 98.6]
(98.6, 196.2]
(196.2, 293.8]
(293.8, 391.4]
(391.4, 489.0]
(489.0, 586.6]
(586.6, 684.2]
(684.2, 781.8]
(781.8, 879.4]
(879.4, 977.0]
cnt_group

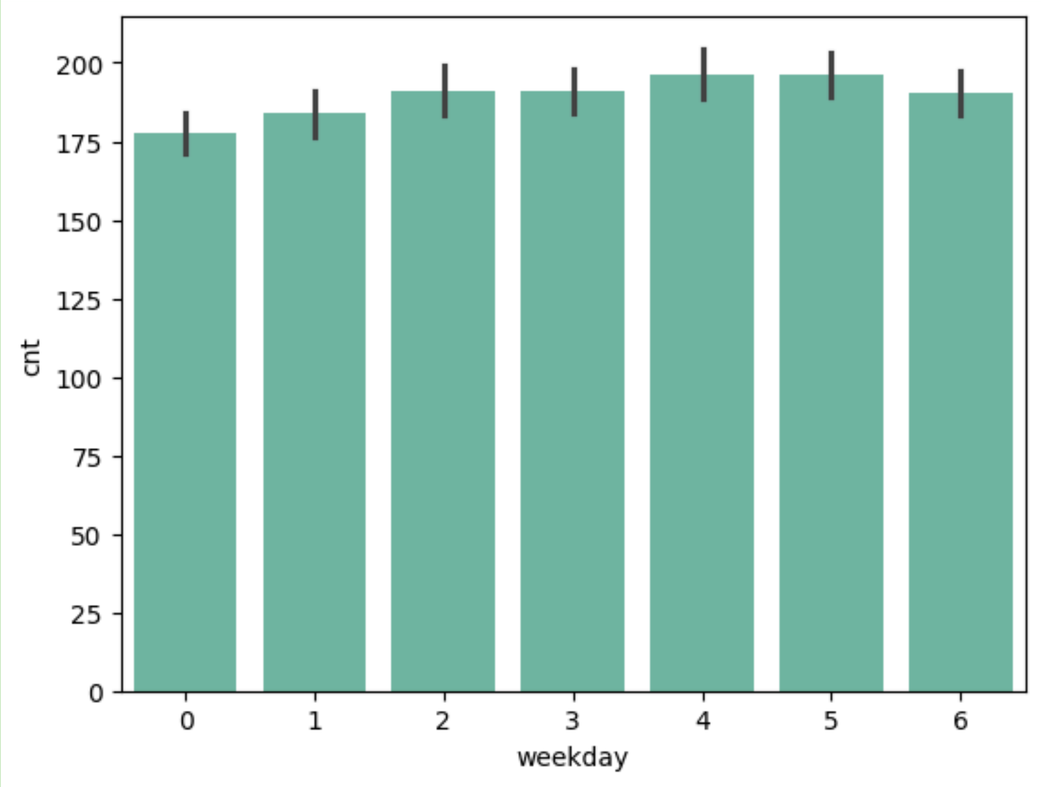
類別變數對cnt, registered,casual的barplot

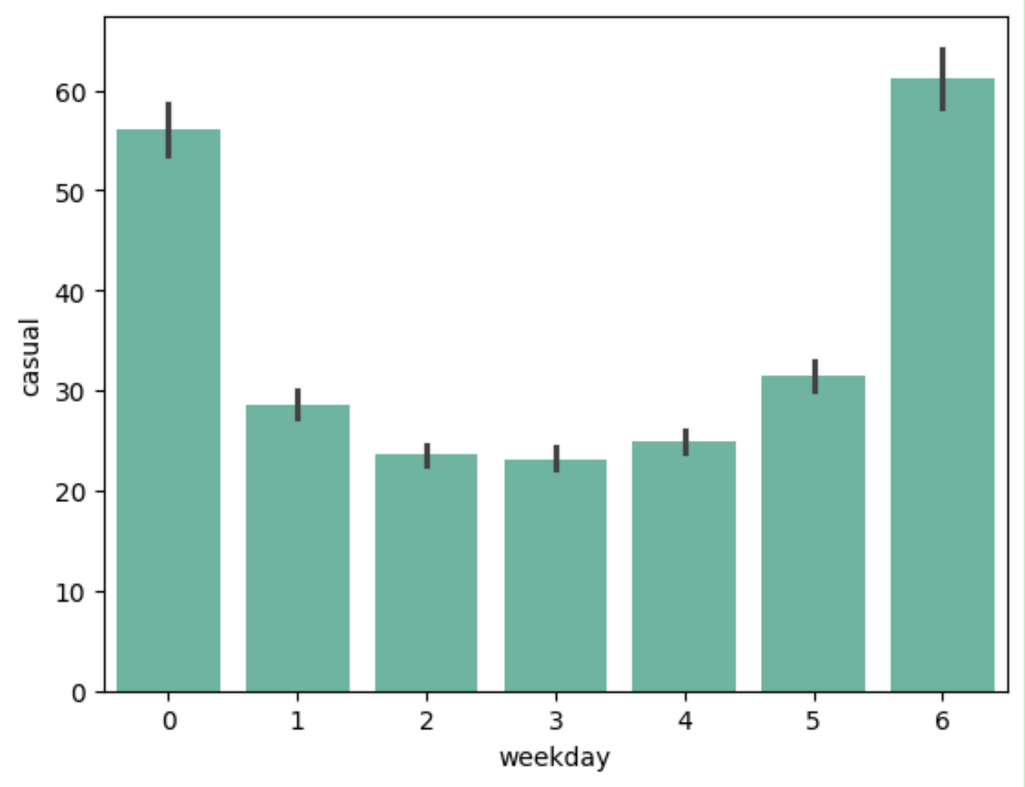


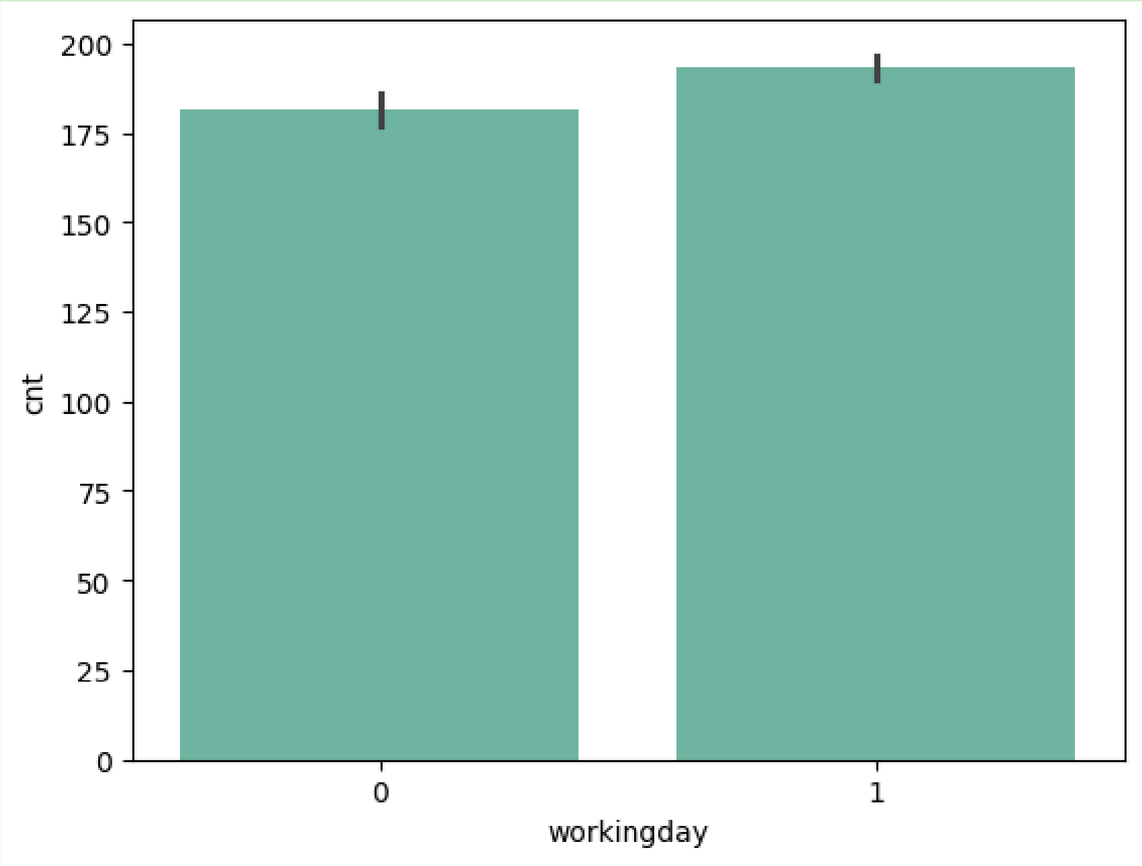


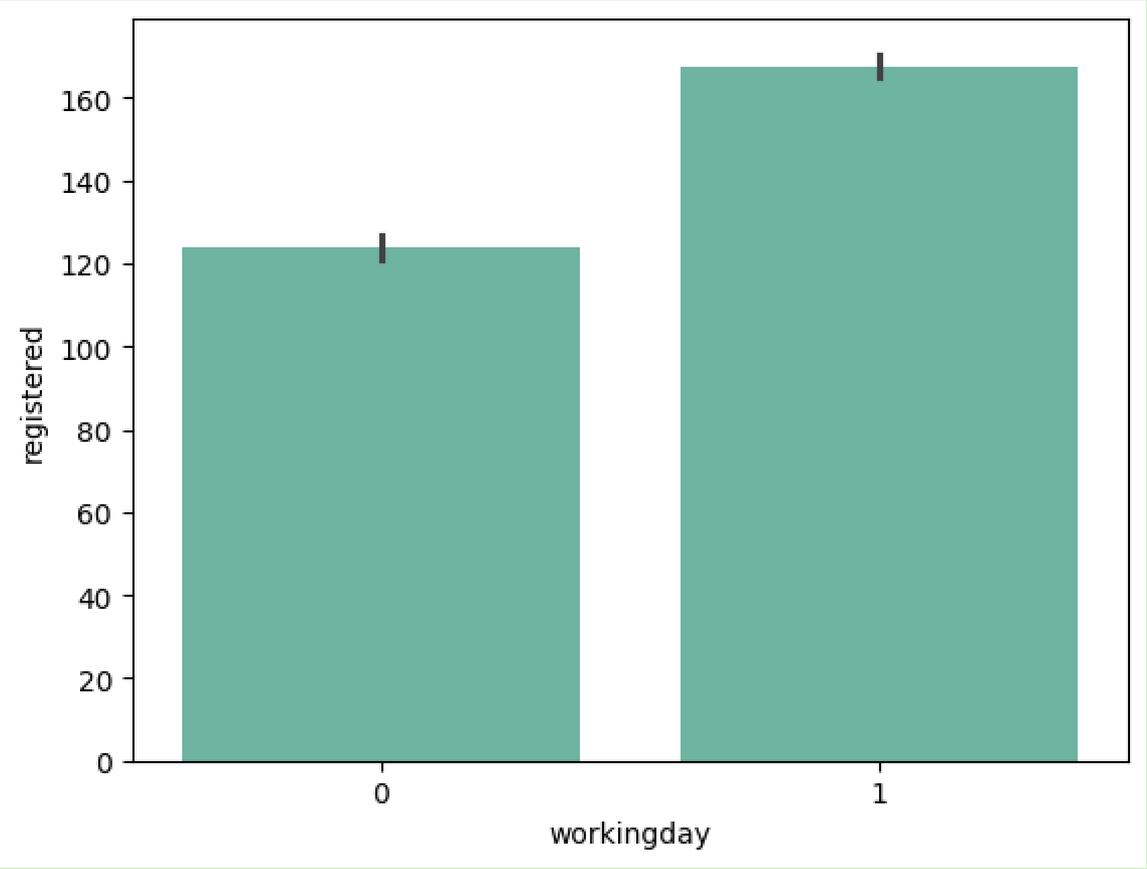

250
200
150
100
50
0
cnt
1
2
3
4
5
6
7
8
9
10
11
12
mnth

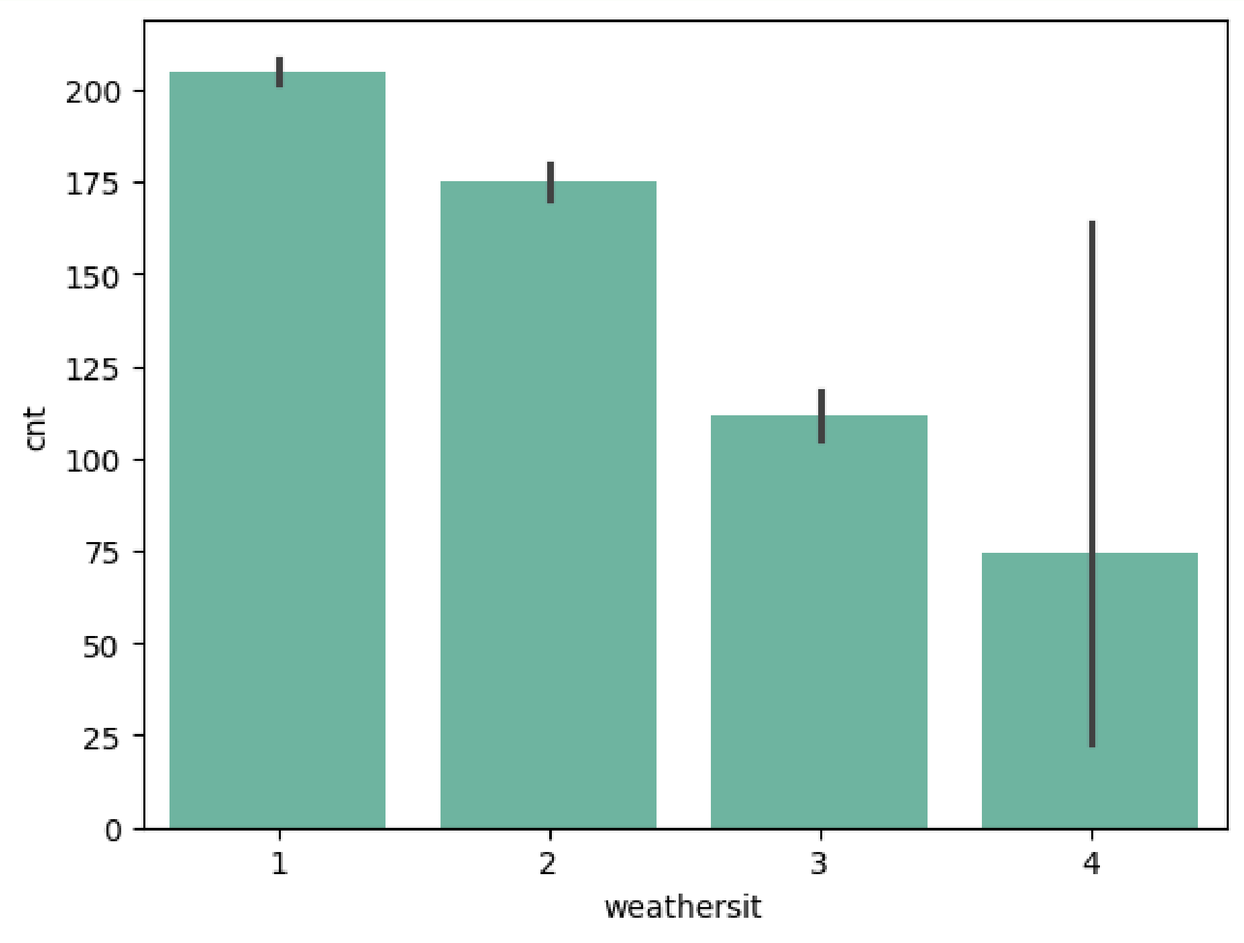

50
40
30
20
10
0
casual
1
2
3
4
5
6
7
8
9
10
11
12
mnth

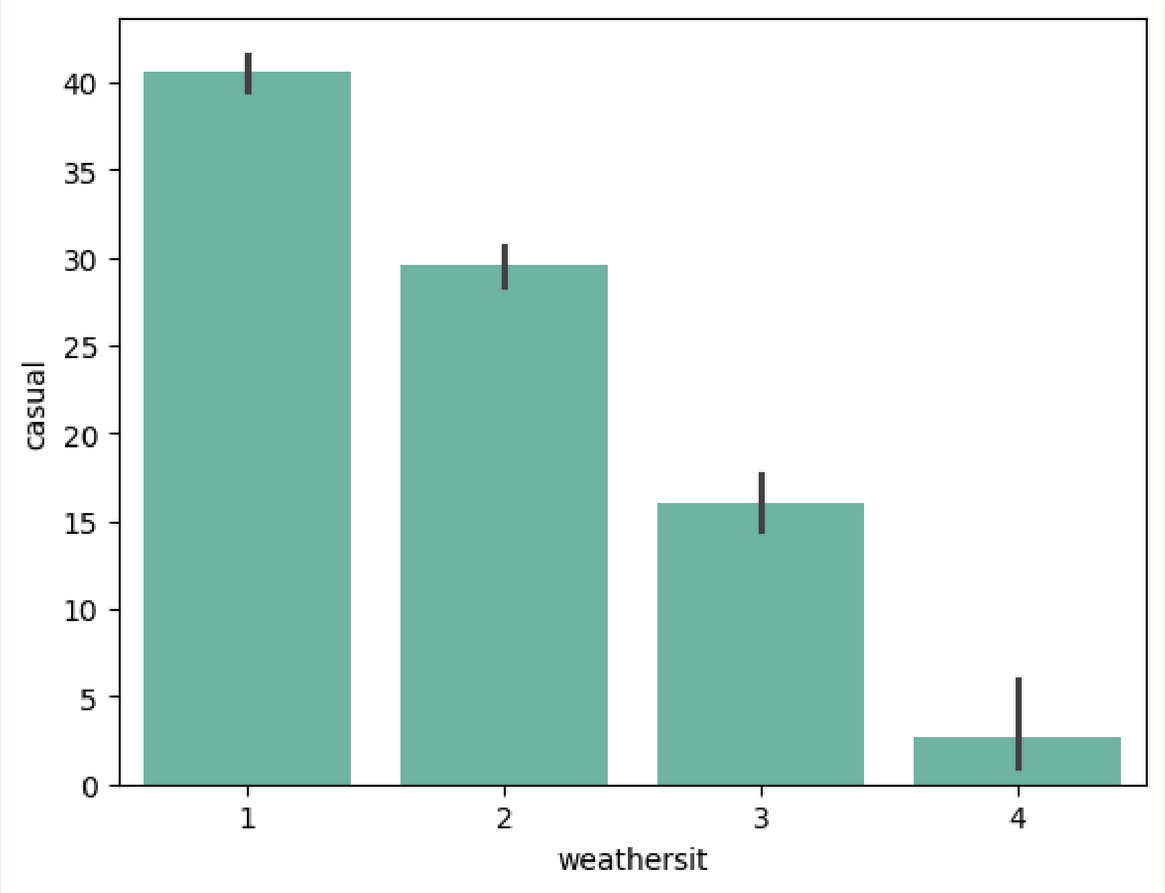

cnt
200
175
150
125
100
75
50
25
0
0
1
2
3
4
5
6
weekday

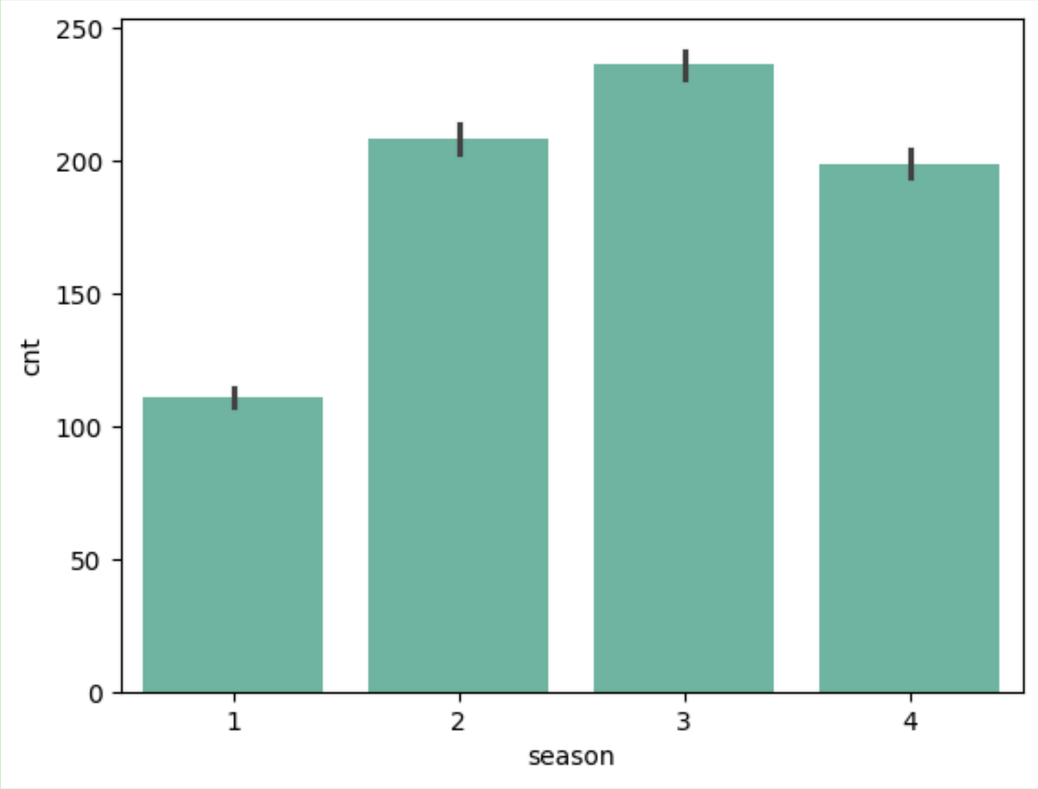

casual
60
50
40
30
20
10
0
0
1
2
3
4
5
6
weekday

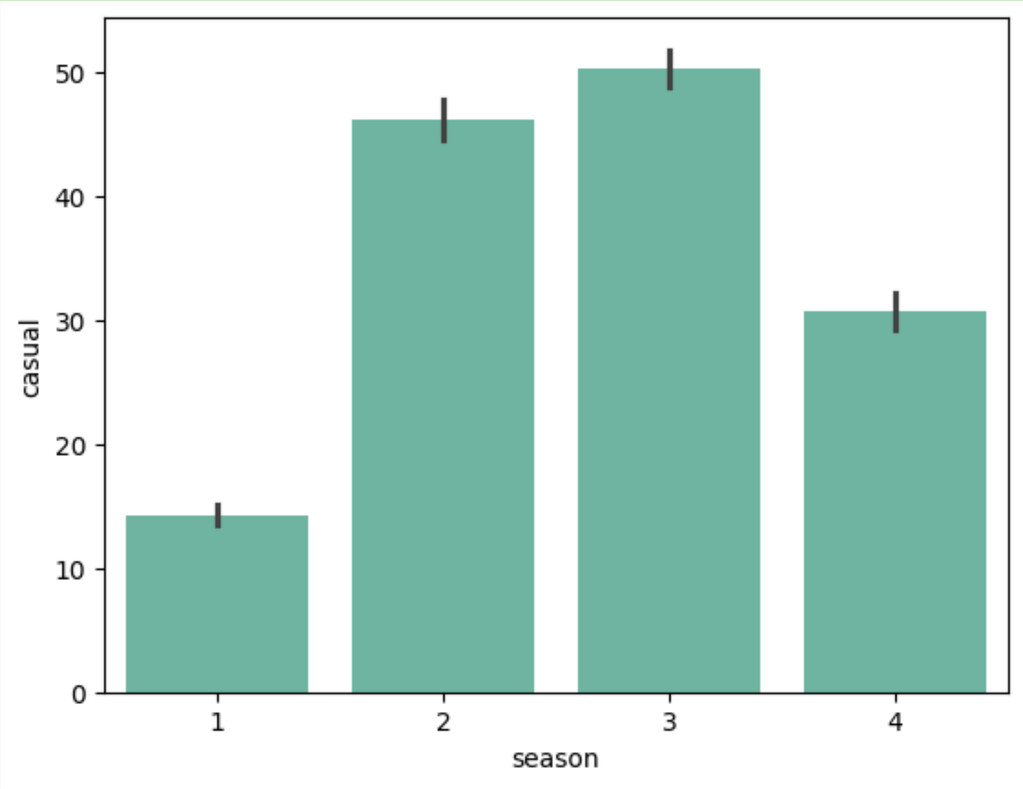

200
175
150
125
100
75
50
25
0
cnt
0
1
workingday

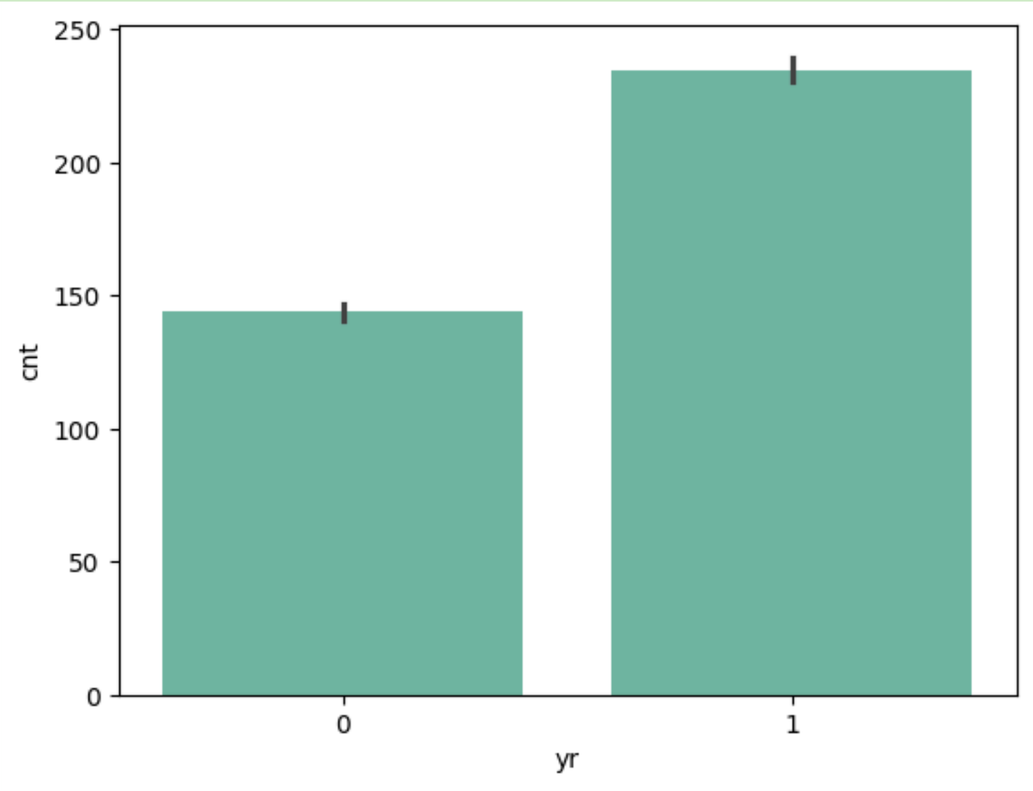

160
140
120
100
80
60
40
20
0
registered
0
1
workingday

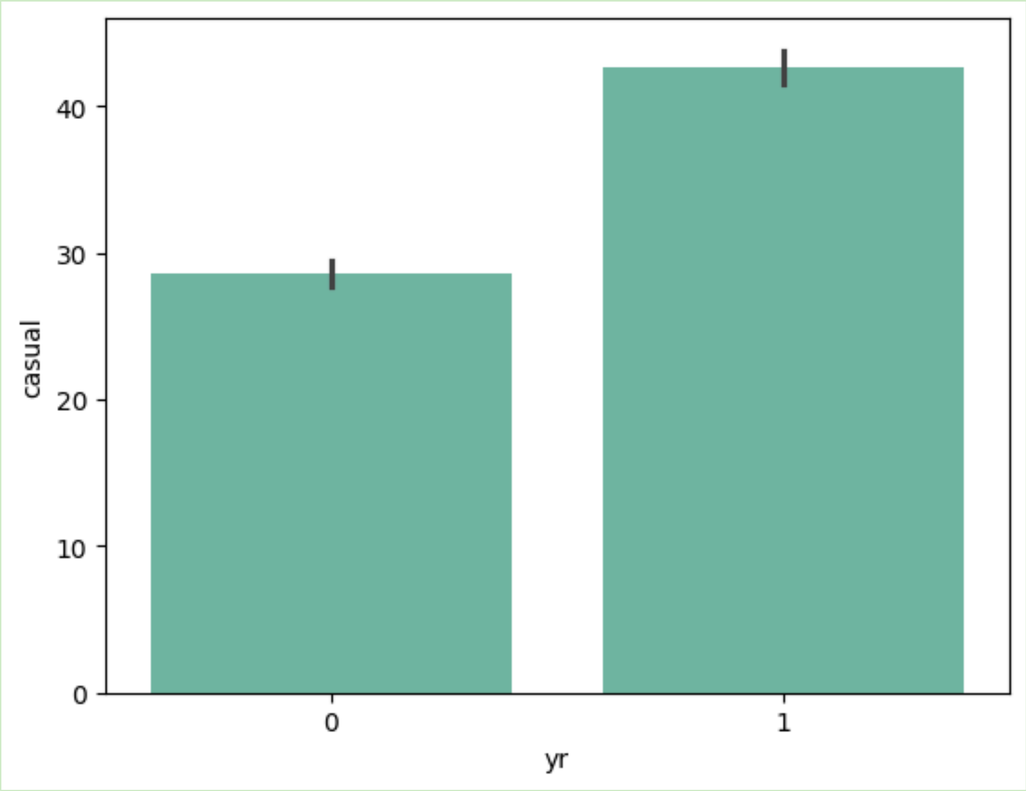

200
175
150
125
100
75
50
25
0
cnt
1
2
3
4
weathersit

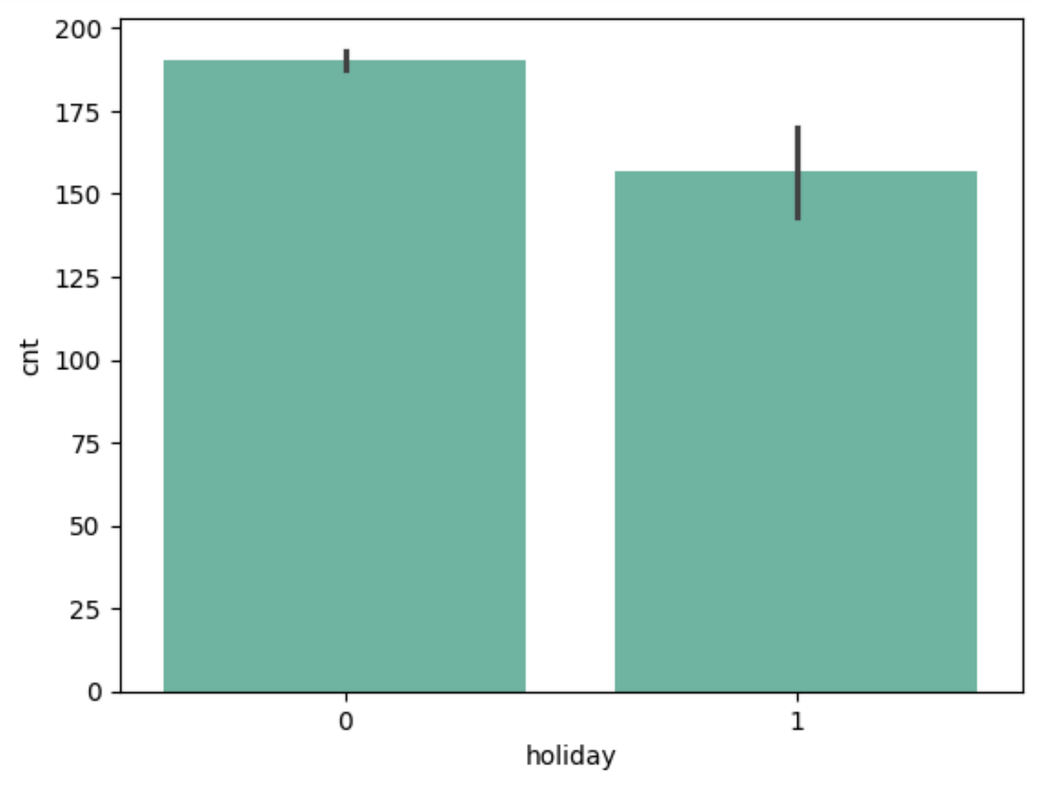

40
35
30
25
20
15
10
5
0
casual
1
2
3
4
weathersit

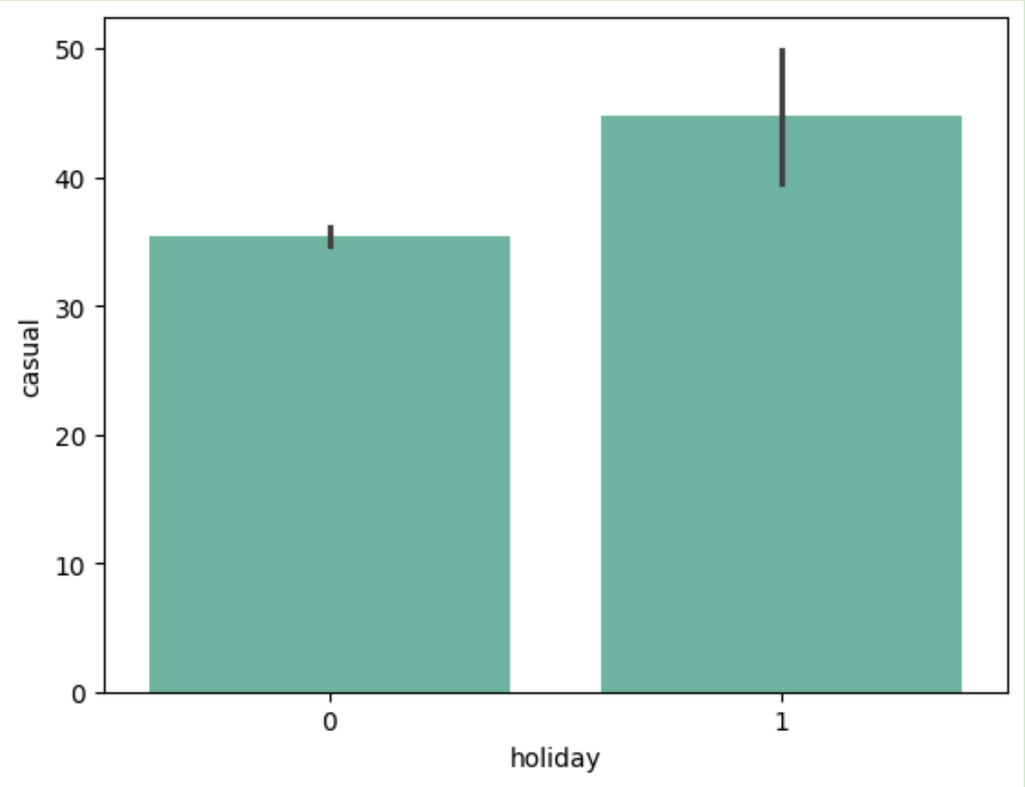

250
200
150
100
50
0
cnt
1
2
3
4
season


50
40
30
20
10
0
casual
1
2
3
4
season

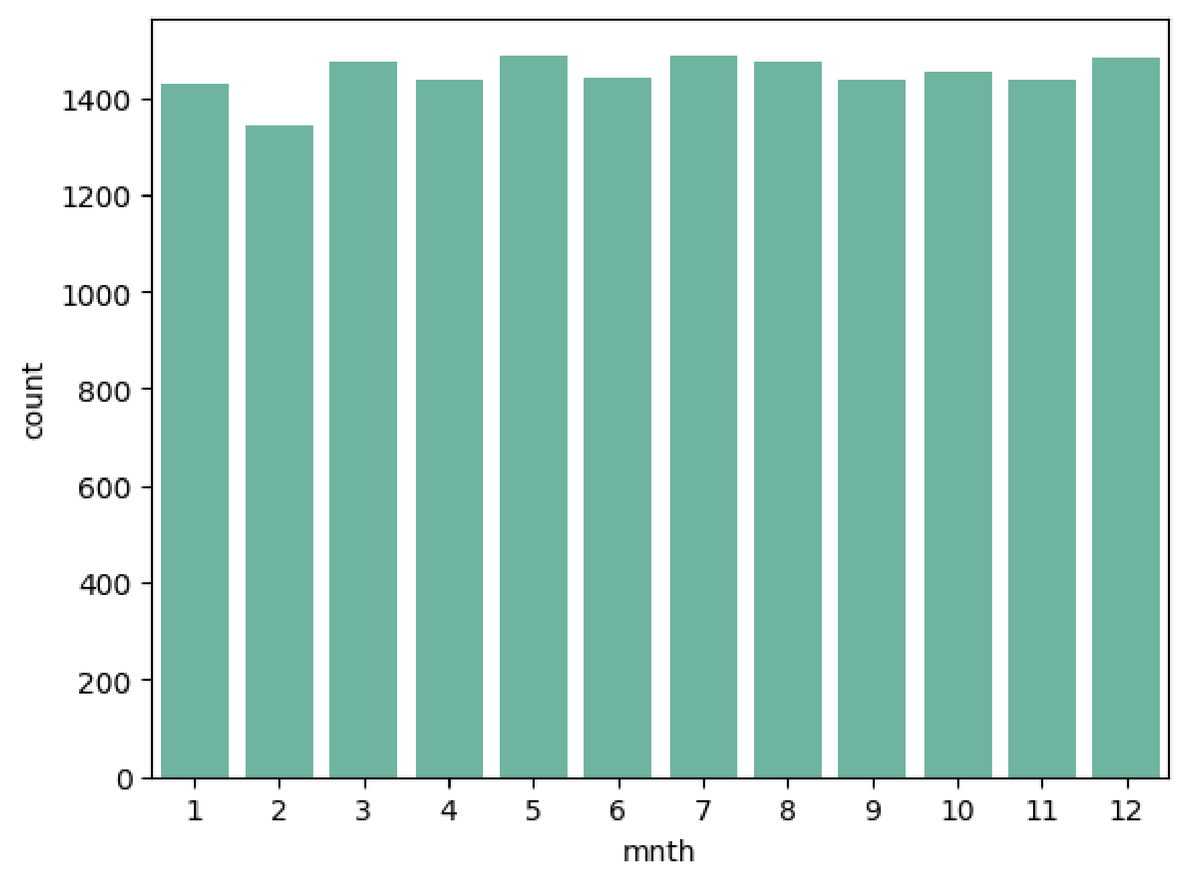

250
200
150
100
50
0
cnt
0
1
yr

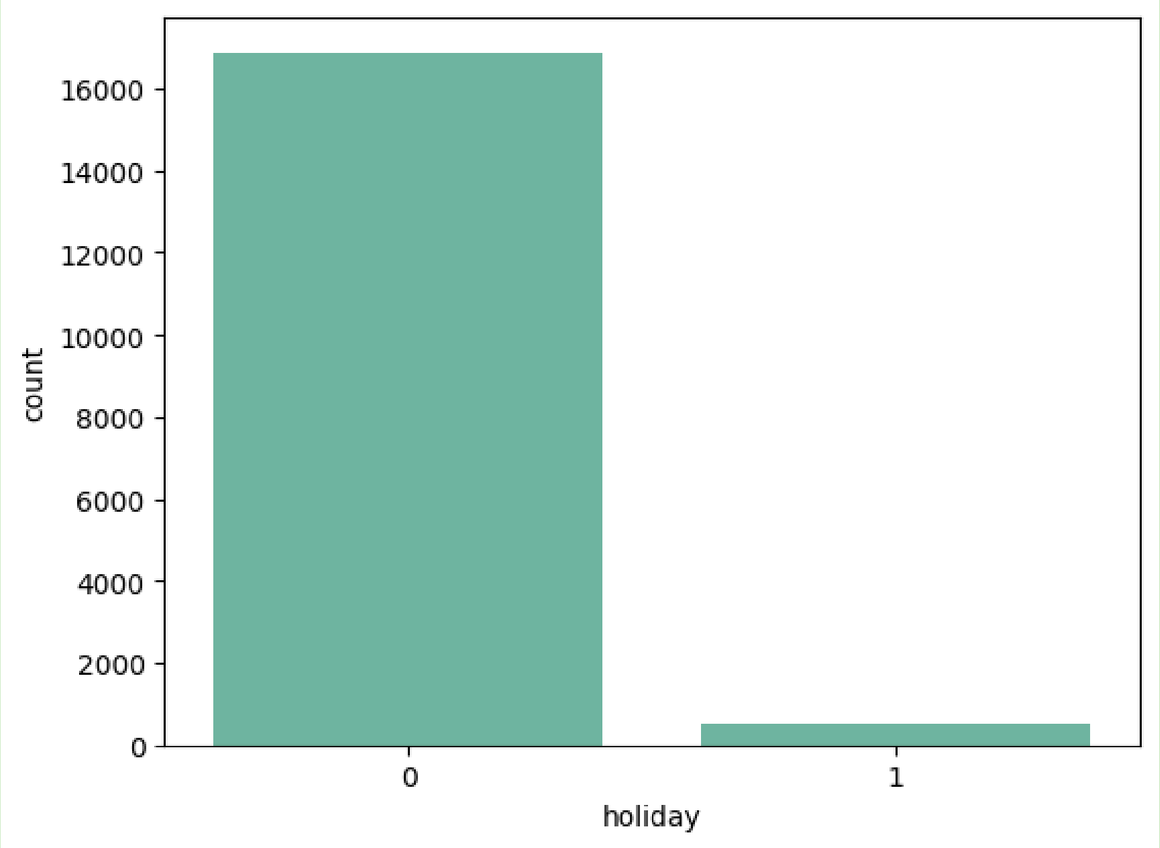

40
30
20
10
0
casual
0
1
yr

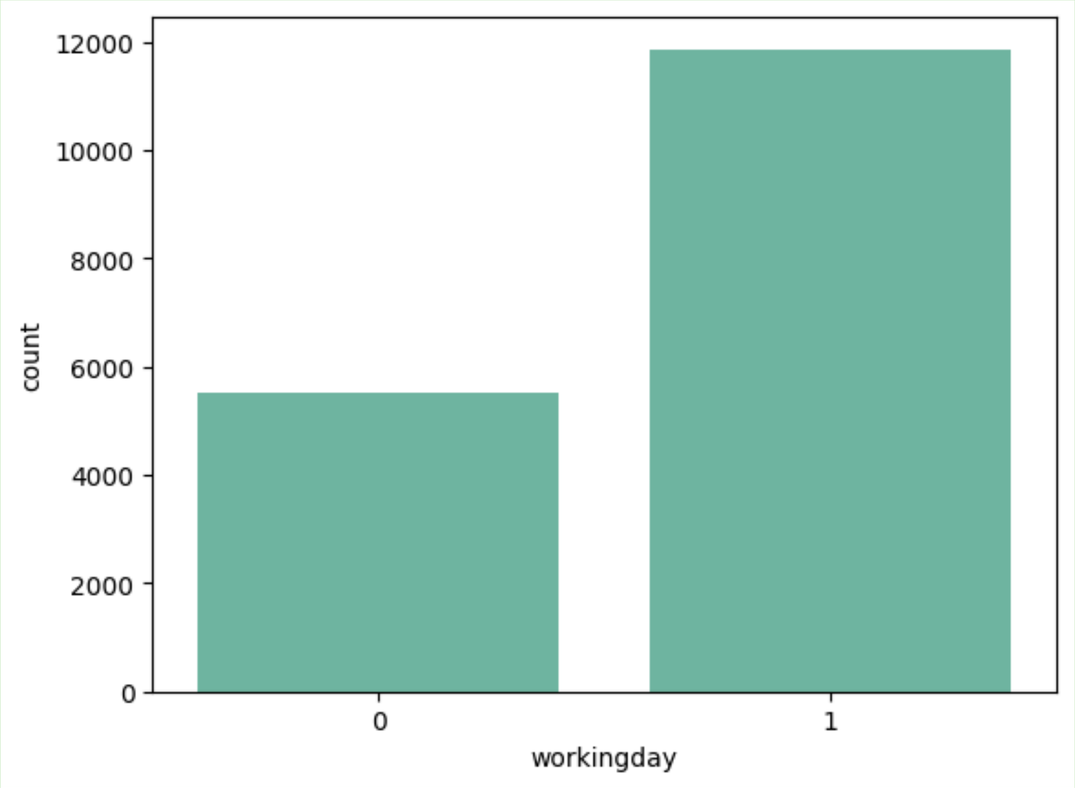

200
175
150
125
100
75
50
25
0
cnt
0
1
holiday

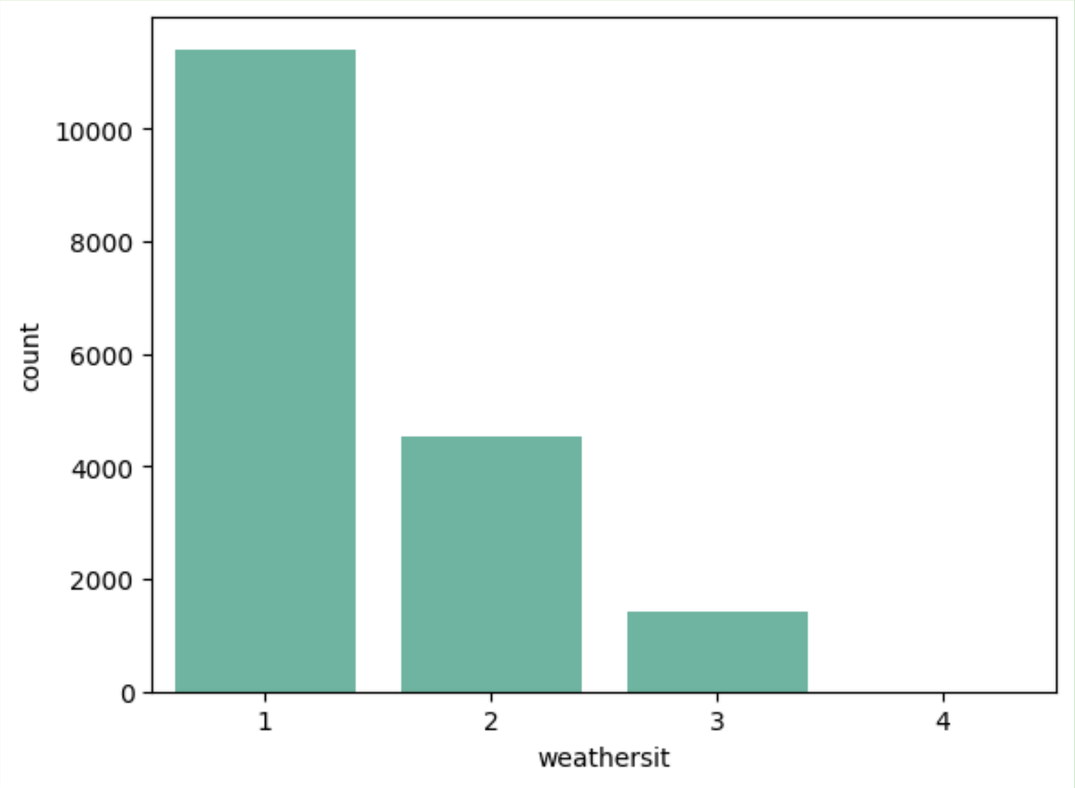

50
40
30
20
10
0
casual
0
1
holiday

類別變數countplot






12000
10000
8000
6000
4000
2000
0
count
0
1
workingday


10000
8000
6000
4000
2000
0
count
1
2
3
4
weathersit

卡方檢定

```
p 值 < 0.05 for season 和 workingday: 0.00012995312553585743
p 值 < 0.05 for season 和 weathersit: 1.1626263980985058e-31
p 值 < 0.05 for mnth 和 season: 0.0
p 值 < 0.05 for mnth 和 weekday: 0.043179498278780 36
p 值 < 0.05 for mnth 和 workingday: 2.7885096954987004e-12
p 值 < 0.05 for mnth 和 weathersit: 1.1892410914534171e-67
p 值 < 0.05 for hr 和 weathersit: 1.4472143076305382e-13
p 值 < 0.05 for holiday 和 season: 9.038993011801841e-08
p 值 < 0.05 for holiday 和 mnth: 8.35161115637833e-44
p 值 < 0.05 for holiday 和 weekday: 5.337493047437229e-301
p 值 < 0.05 for holiday 和 workingday: 3.429444004362583e-242
p 值 < 0.05 for holiday 和 weathersit: 0.02000137417993775
p 值 < 0.05 for weekday 和 workingday: 0.0
p 值 < 0.05 for weathersit 和 yr: 0.000403540945568836
p 值 < 0.05 for weathersit 和 weekday: 4.507973791100056e-16
p 值 < 0.05 for weathersit 和 workingday: 1.3151438111663914e-07
```

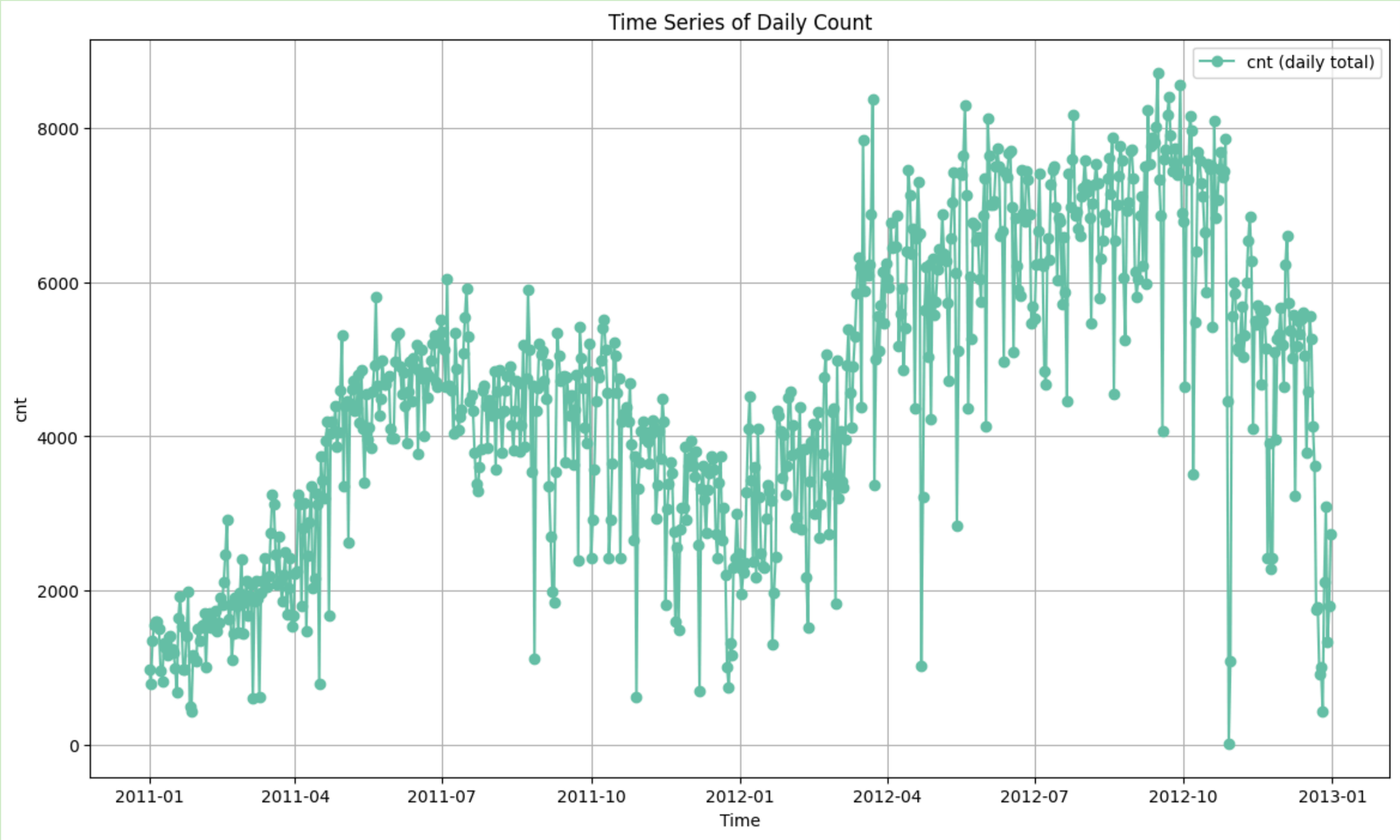
Outliers

异常值比例: 14.85%

	season	yr	mnth	hr	holiday	weekday	workingday	weathersit	temp	atemp	hum	windspeed	casual	registered	cnt
175	1	0	1	14	0	6	0	1	0.20	0.1667	0.32	0.4925	12	83	95
178	1	0	1	17	0	6	0	1	0.16	0.1212	0.37	0.5522	5	64	69
194	1	0	1	9	0	0	0	1	0.12	0.0758	0.46	0.5224	0	19	19
196	1	0	1	11	0	0	0	1	0.16	0.1212	0.40	0.5224	2	47	49
265	1	0	1	12	0	3	1	1	0.20	0.1515	0.47	0.5821	3	52	55
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
17327	1	1	12	20	0	6	0	2	0.30	0.2424	0.56	0.5522	2	64	66
17328	1	1	12	21	0	6	0	2	0.28	0.2424	0.56	0.4925	4	56	60
17341	1	1	12	10	0	0	0	1	0.26	0.2121	0.41	0.5821	31	91	122
17343	1	1	12	12	0	0	0	1	0.28	0.2273	0.36	0.5821	47	97	144
17344	1	1	12	13	0	0	0	1	0.30	0.2576	0.36	0.6567	49	120	169

2580 rows × 15 columns

時間序列


Time Series of Daily Count
cnt (daily total)
cnt
8000
6000
4000
2000
0
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
2013-01
Time

EDA總結

1. cnt, registered, casual, hum, windspeed, weathersit, holiday皆有離群值，但weathersit與hum的偏少
2. 可以看出atemp與temp、registered與cnt有強線性關係。casual與cnt可能有線性關係，temp與casual、atemp與casual可能有強非線性關係。temp有雙峰趨勢，atemp與hum有右偏、windspeed有左偏情形。casual, registered, cnt皆為右偏。
3. temp與atemp、registed與目標cnt具有共線性，考慮先將前者丟棄其一(temp)去建模
4. 考慮將有顯著統計關系的類別變數之間考慮生成交互項
5. 季節因素影響是巨大的

建模策略與特徵工程

1.移除高共線性特徵(temp)、registered、casual

2. 確認cnt是否0膨脹與期望值、變異數

3.將outliers做capping

4.數值類變數考慮進行標準化

5.建模

6.如果是迴歸模型檢查殘差與標準化殘差分布

7.異常處理與重新建模:(train,valid)=(0.8,0.2)

發現cnt 0膨脹比例: 0 %

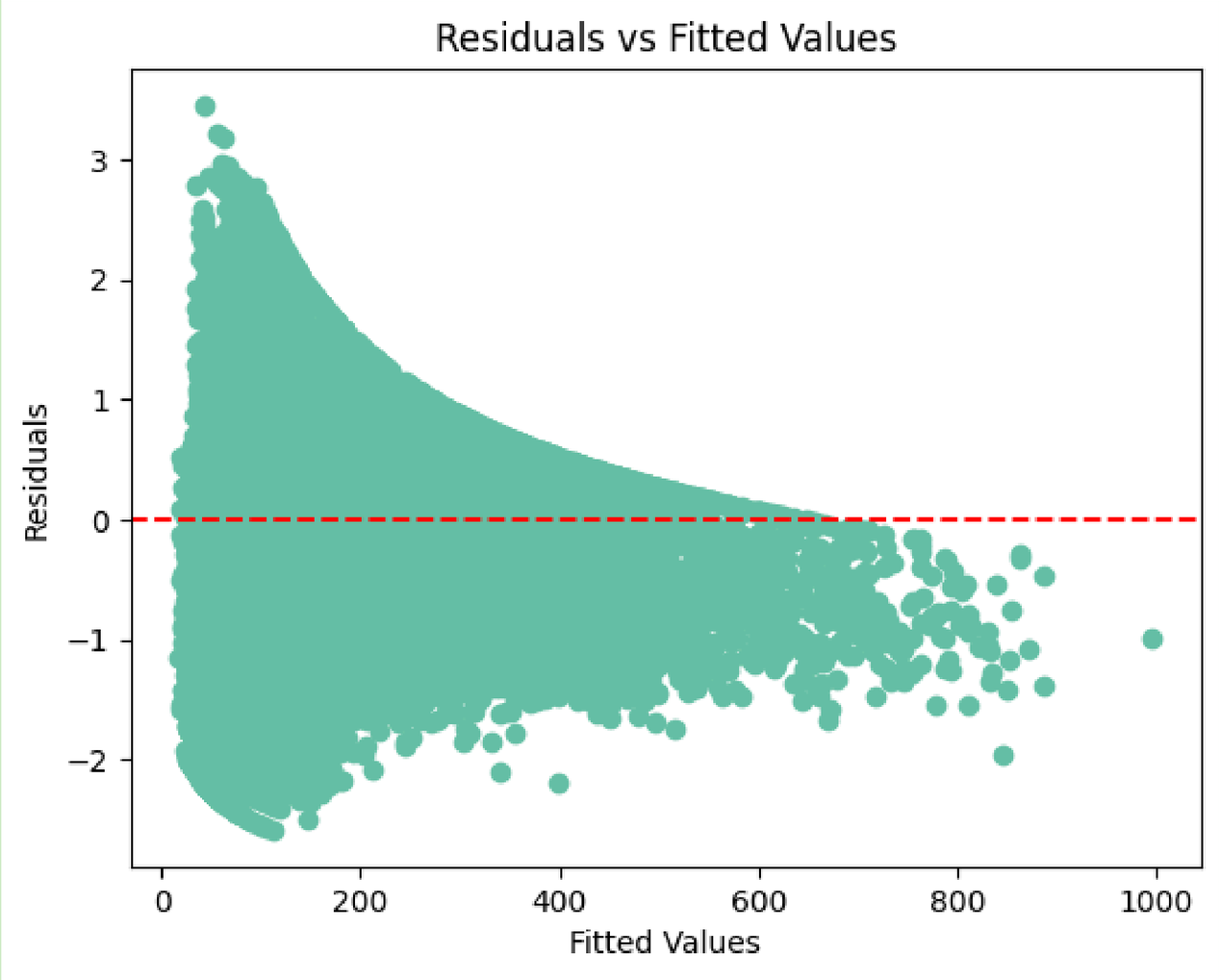
期望值: 189.463,

變異數: 32901.461,

所以使用NegBin Reg model建模

原始建模

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                    cnt   No. Observations:                17379
Model:                            GLM   Df Residuals:                    17367
Model Family:        NegativeBinomial   Df Model:                           11
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:            -1.0442e+05
Date:                Wed, 04 Dec 2024   Deviance:                       15869.
Time:                        09:39:08   Pearson chi2:                 1.53e+04
No. Iterations:                     15   Pseudo R-squ. (CS):             0.3584
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      3.2088      0.049     65.449      0.000       3.113       3.305
season         0.1438      0.013     11.149      0.000       0.118       0.169
yr             0.4637      0.015     30.256      0.000       0.434       0.494
mnth          -0.0029      0.004     -0.727      0.467      -0.011       0.005
hr             0.0728      0.001     62.309      0.000       0.070       0.075
holiday       -0.1517      0.047     -3.198      0.001      -0.245      -0.059
weekday        0.0089      0.004      2.321      0.020       0.001       0.016
workingday     0.0738      0.017      4.349      0.000       0.041       0.107
weathersit    -0.0457      0.013     -3.393      0.001      -0.072      -0.019
atemp          1.7519      0.048     36.516      0.000       1.658       1.846
hum           -0.8058      0.049    -16.552      0.000      -0.901      -0.710
windspeed      0.2448      0.069      3.556      0.000       0.110       0.380
==============================================================================
```

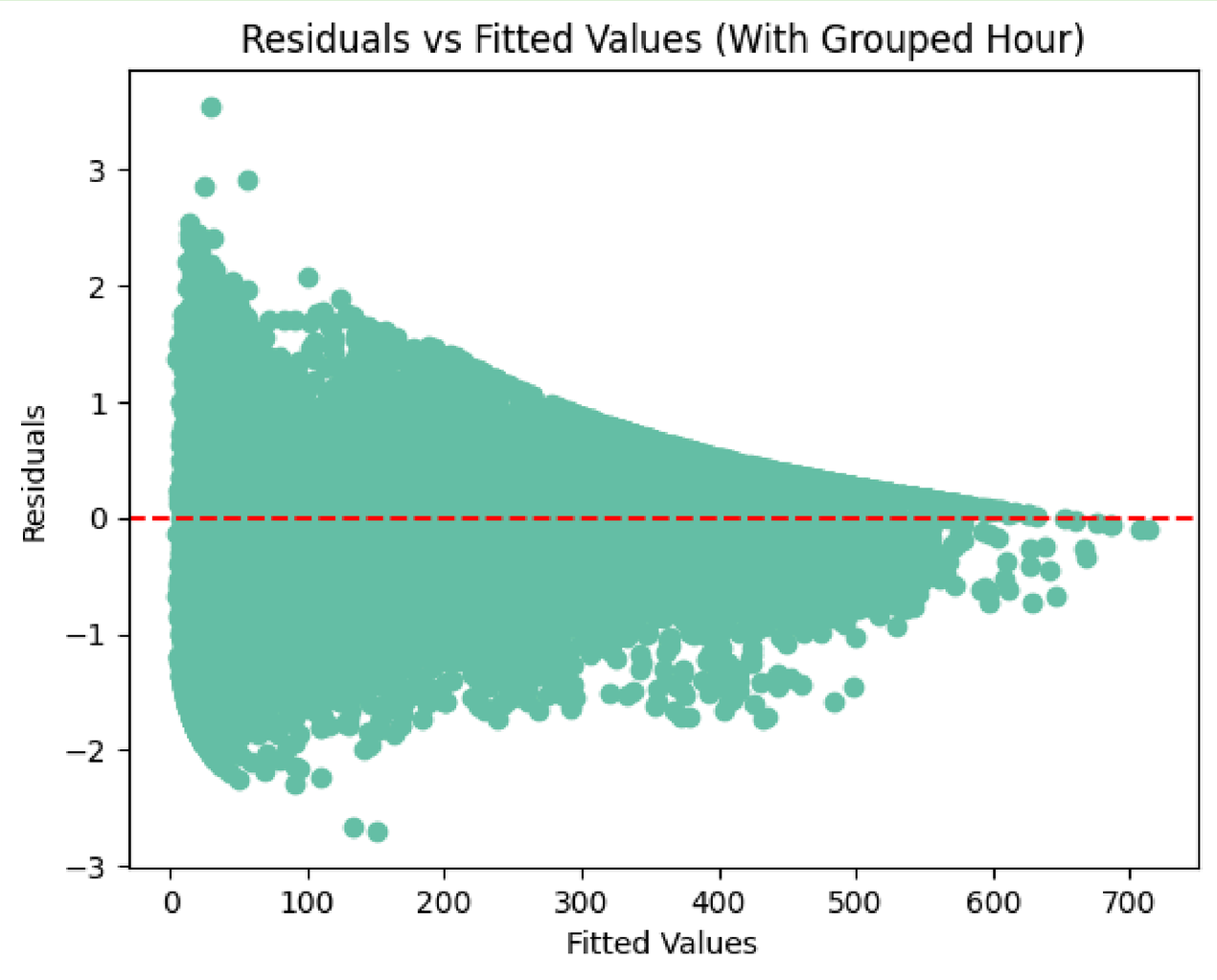

Residuals vs Fitted Values
Residuals
Fitted Values
3
2
1
0
−1
−2
0
200
400
600
800
1000

Generalized Linear Model Regression Results

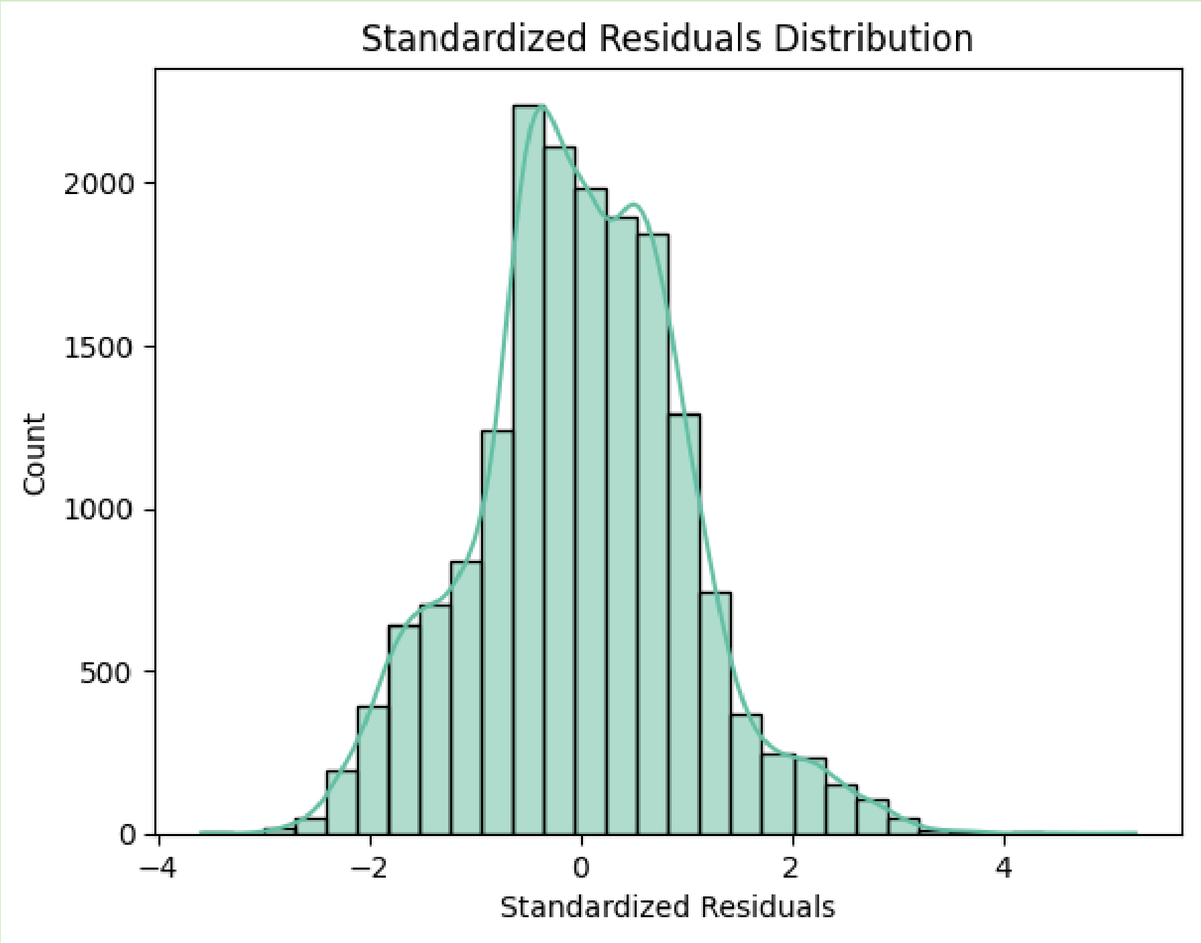
Dep. Variable:	cnt	No. Observations:	17379
Model:	GLM	Df Residuals:	17342
Model Family:	NegativeBinomial	Df Model:	36
Link Function:	Log	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-1.0104e+05
Date:	Wed, 04 Dec 2024	Deviance:	9113.2
Time:	09:39:22	Pearson chi2:	9.16e+03
No. Iterations:	11	Pseudo R-squ. (CS):	0.5650
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	1.6646	0.103	16.204	0.000	1.463	1.866
C(season)[T.2]	0.2520	0.048	5.232	0.000	0.158	0.346
C(season)[T.3]	0.3635	0.057	6.379	0.000	0.252	0.475
C(season)[T.4]	0.5705	0.049	11.746	0.000	0.475	0.666
C(yr)[T.1]	0.4552	0.016	29.333	0.000	0.425	0.486
C(mnth)[T.2]	0.0621	0.039	1.572	0.116	-0.015	0.139
C(mnth)[T.3]	0.0692	0.045	1.549	0.121	-0.018	0.157
C(mnth)[T.4]	0.0213	0.066	0.325	0.745	-0.107	0.150
C(mnth)[T.5]	0.1848	0.068	2.698	0.007	0.051	0.319
C(mnth)[T.6]	0.1529	0.069	2.214	0.027	0.018	0.288
C(mnth)[T.7]	0.0791	0.078	1.015	0.310	-0.074	0.232
C(mnth)[T.8]	0.0675	0.076	0.892	0.373	-0.081	0.216
C(mnth)[T.9]	0.0745	0.069	1.084	0.278	-0.060	0.209
C(mnth)[T.10]	-0.0863	0.065	-1.328	0.184	-0.214	0.041
C(mnth)[T.11]	-0.1491	0.063	-2.369	0.018	-0.272	-0.026
C(mnth)[T.12]	-0.1200	0.050	-2.402	0.016	-0.218	-0.022
C(hr_grouped)[T.Morning]	1.8776	0.024	78.402	0.000	1.831	1.925
C(hr_grouped)[T.Afternoon]	1.9193	0.032	60.648	0.000	1.857	1.981
C(hr_grouped)[T.Evening]	1.3086	0.036	36.644	0.000	1.239	1.379
C(holiday)[T.1]	0.0390	0.069	0.562	0.574	-0.097	0.175
C(weekday)[T.1]	-0.0904	0.020	-4.473	0.000	-0.130	-0.051
C(weekday)[T.2]	-0.0775	0.023	-3.358	0.001	-0.123	-0.032
C(weekday)[T.3]	-0.0490	0.023	-2.115	0.034	-0.094	-0.004
C(weekday)[T.4]	-0.0339	0.023	-1.446	0.148	-0.080	0.012
C(weekday)[T.5]	0.0013	0.024	0.055	0.956	-0.045	0.048
C(weekday)[T.6]	0.0442	0.028	1.552	0.121	-0.012	0.100
C(workingday)[T.1]	-0.2884	0.030	-9.610	0.000	-0.347	-0.230
C(weathersit)[T.2]	-0.0550	0.019	-2.898	0.004	-0.092	-0.018
C(weathersit)[T.3]	-0.4710	0.033	-14.350	0.000	-0.535	-0.407
C(weathersit)[T.4]	-0.0913	0.587	-0.156	0.876	-1.241	1.058
atemp	4.2417	0.291	14.563	0.000	3.671	4.813
atemp_squared	-3.2100	0.308	-10.433	0.000	-3.813	-2.607
hum	0.9688	0.255	3.800	0.000	0.469	1.469
hum_squared	-1.0181	0.205	-4.978	0.000	-1.419	-0.617
windspeed	0.5519	0.195	2.836	0.005	0.171	0.933
windspeed_squared	-1.3316	0.438	-3.043	0.002	-2.189	-0.474
hr_workingday	0.0298	0.002	13.316	0.000	0.025	0.034
holiday_weekday	-0.0622	0.034	-1.855	0.064	-0.128	0.004

將類別特徵重編碼加上交互項與變換建模

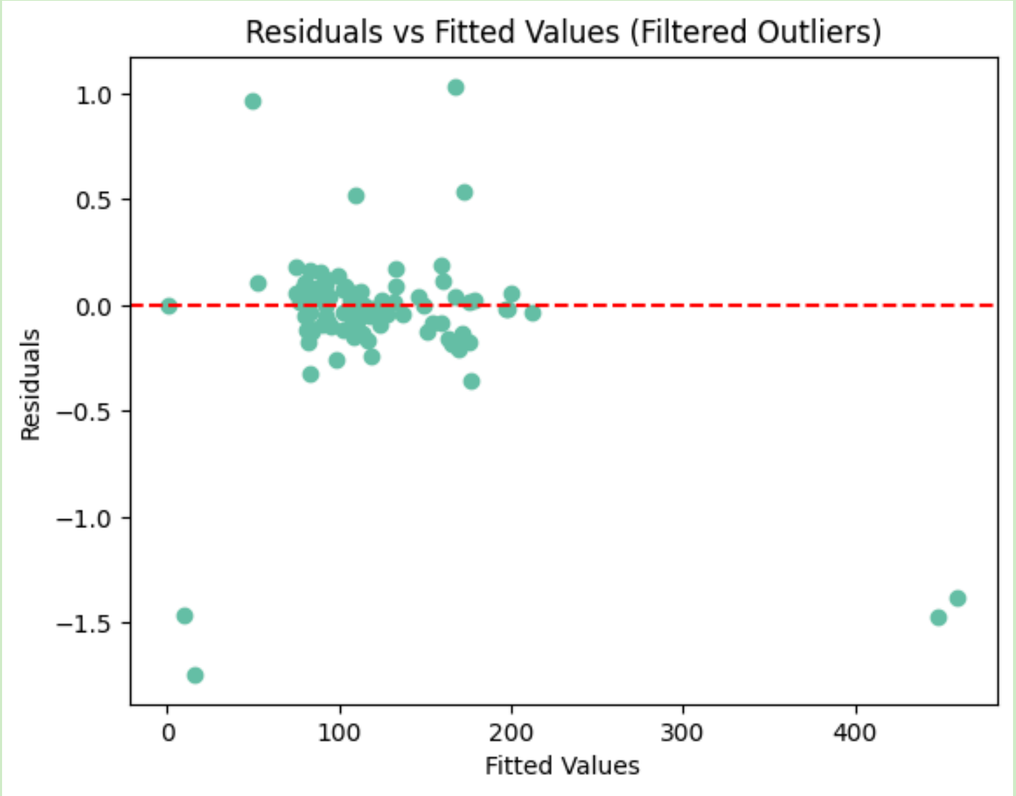

Residuals vs Fitted Values (With Grouped Hour)
Residuals
Fitted Values
3
2
1
0
−1
−2
−3
0
100
200
300
400
500
600
700

檢查異常值



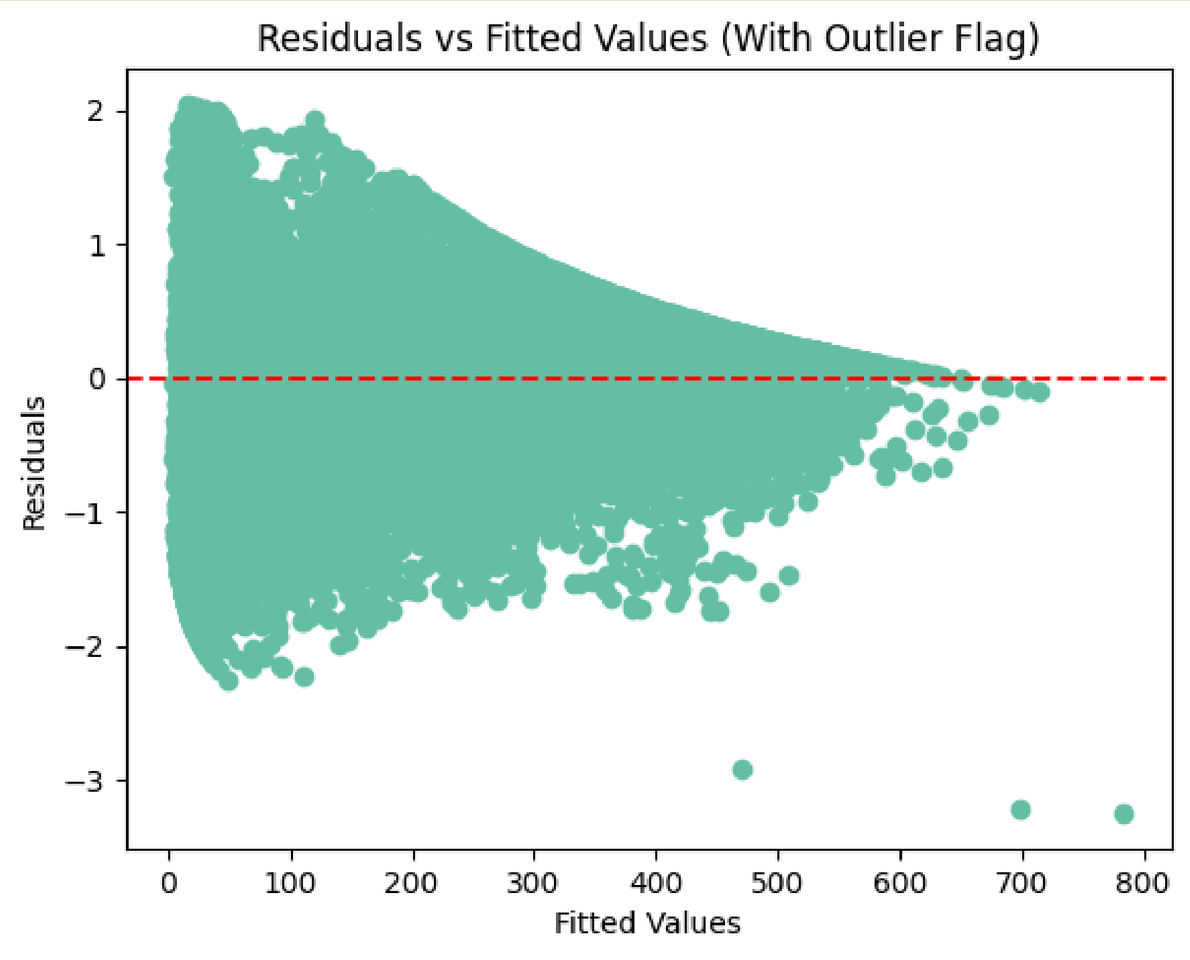
異常值筆數:56

將異常值與正常資料分別建模觀察殘差



```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                    cnt   No. Observations:               17379
Model:                            GLM   Df Residuals:                   17341
Model Family:        NegativeBinomial   Df Model:                          37
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:            -1.0091e+05
Date:                Wed, 04 Dec 2024   Deviance:                       8854.7
Time:                        10:01:33   Pearson chi2:                 8.73e+03
No. Iterations:                     11   Pseudo R-squ. (CS):             0.5715
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                  1.5477      0.103     15.048      0.000       1.346       1.749
C(season)[T.2]             0.2677      0.048      5.556      0.000       0.173       0.362
C(season)[T.3]             0.3888      0.057      6.821      0.000       0.277       0.501
C(season)[T.4]             0.5868      0.049     12.077      0.000       0.492       0.682
C(yr)[T.1]                 0.4513      0.016     29.076      0.000       0.421       0.482
C(mnth)[T.2]               0.0735      0.040      1.859      0.063      -0.004       0.151
C(mnth)[T.3]               0.0718      0.045      1.607      0.108      -0.016       0.159
C(mnth)[T.4]               0.0193      0.066      0.294      0.768      -0.109       0.148
C(mnth)[T.5]               0.1786      0.069      2.608      0.009       0.044       0.313
C(mnth)[T.6]               0.1462      0.069      2.117      0.034       0.011       0.282
C(mnth)[T.7]               0.0804      0.078      1.031      0.302      -0.072       0.233
C(mnth)[T.8]               0.0603      0.076      0.796      0.426      -0.088       0.209
C(mnth)[T.9]               0.0593      0.069      0.863      0.388      -0.075       0.194
C(mnth)[T.10]             -0.0967      0.065     -1.488      0.137      -0.224       0.031
C(mnth)[T.11]             -0.1756      0.063     -2.789      0.005      -0.299      -0.052
C(mnth)[T.12]             -0.1352      0.050     -2.705      0.007      -0.233      -0.037
C(hr_grouped)[T.Morning]   1.9171      0.024     79.921      0.000       1.870       1.964
C(hr_grouped)[T.Afternoon] 1.9525      0.032     61.623      0.000       1.890       2.015
C(hr_grouped)[T.Evening]   1.3392      0.036     37.463      0.000       1.269       1.409
C(holiday)[T.1]            0.0487      0.069      0.702      0.483      -0.087       0.185
C(weekday)[T.1]           -0.0939      0.020     -4.647      0.000      -0.134      -0.054
C(weekday)[T.2]           -0.0873      0.023     -3.784      0.000      -0.133      -0.042
C(weekday)[T.3]           -0.0614      0.023     -2.652      0.008      -0.107      -0.016
C(weekday)[T.4]           -0.0344      0.023     -1.467      0.142      -0.080       0.012
C(weekday)[T.5]            0.0120      0.024      0.506      0.613      -0.034       0.058
C(weekday)[T.6]            0.0432      0.028      1.515      0.130      -0.013       0.099
C(workingday)[T.1]        -0.3138      0.030    -10.451      0.000      -0.373      -0.255
C(weathersit)[T.2]        -0.0538      0.019     -2.836      0.005      -0.091      -0.017
C(weathersit)[T.3]        -0.4939      0.033    -15.037      0.000      -0.558      -0.430
C(weathersit)[T.4]        -0.0437      0.587     -0.074      0.941      -1.194       1.106
atemp                      4.5058      0.292     15.456      0.000       3.934       5.077
atemp_squared             -3.4845      0.308    -11.316      0.000      -4.088      -2.881
hum                        1.0750      0.255      4.215      0.000       0.575       1.575
hum_squared               -1.1129      0.205     -5.440      0.000      -1.514      -0.712
windspeed                  0.5815      0.195      2.988      0.003       0.200       0.963
windspeed_squared         -1.3344      0.438     -3.048      0.002      -2.192      -0.476
hr_workingday              0.0307      0.002     13.736      0.000       0.026       0.035
holiday_weekday           -0.0653      0.034     -1.947      0.052      -0.131       0.000
is_outlier                 1.6424      0.135     12.138      0.000       1.377       1.908
==========================================================================================
```

將是否是
outlier當作變
數重新建模

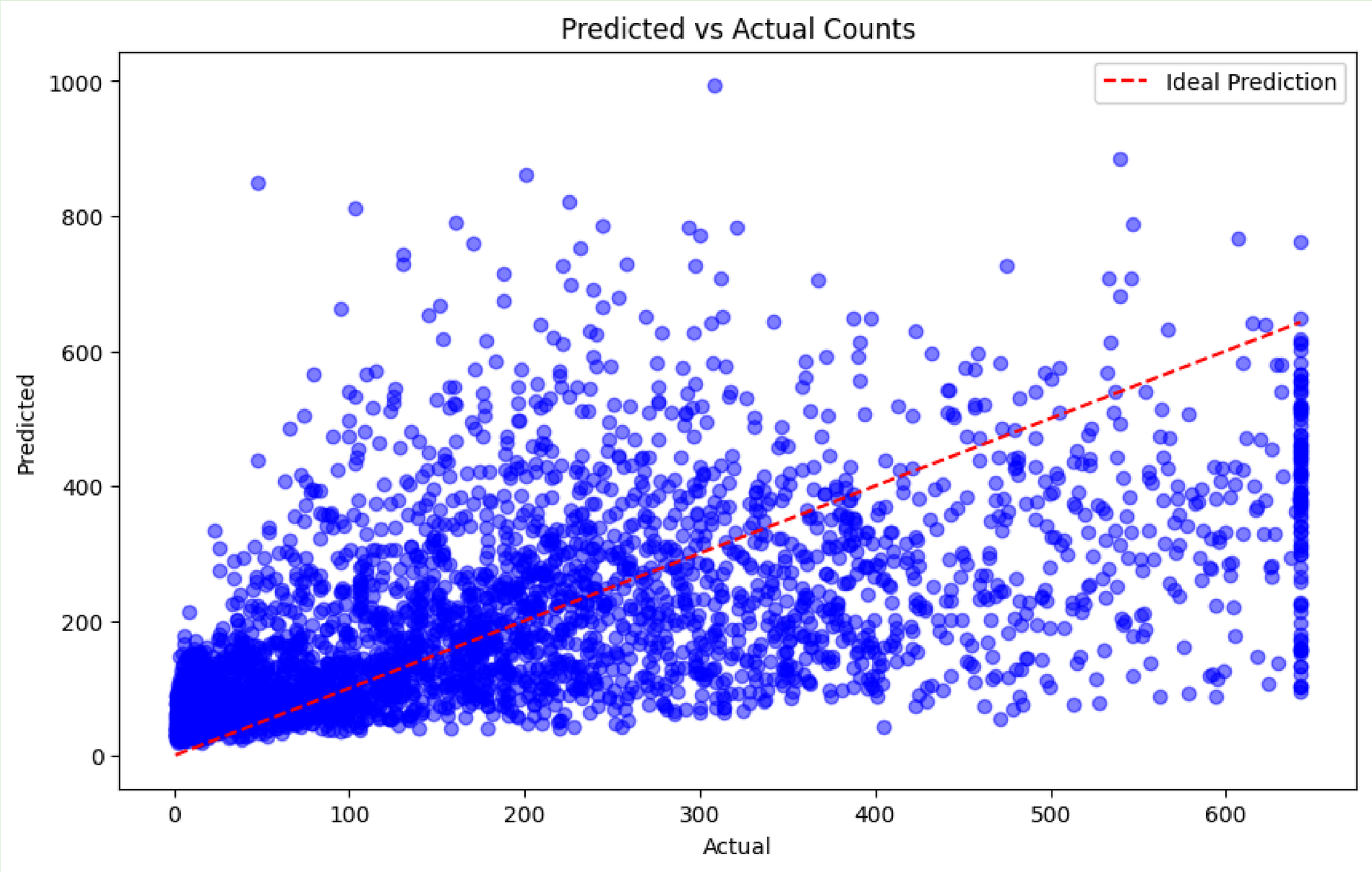

Residuals vs Fitted Values (With Outlier Flag)
Residuals
Fitted Values
2
1
0
−1
−2
−3
0
100
200
300
400
500
600
700
800

結論

1.選擇使用pseudo R^2較高的最後一個模型

2. pseudo R^2=57.15 %

3. 秋季、2012、5月、下午、星期日相較星期一到三、非工作日、無雲天氣相較少雲與多雲、體感溫度越高、濕度越高、風速越低與異常值個體的共享單車每小時使用量是比較高的


Predicted vs Actual Counts
Ideal Prediction
Predicted
Actual
0
200
400
600
800
1000
0
100
200
300
400
500
600

Reference

UCI Machine Learning Repository

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THANKYOU